

Microscopie à force atomique (AFM)

Sandor Kasas

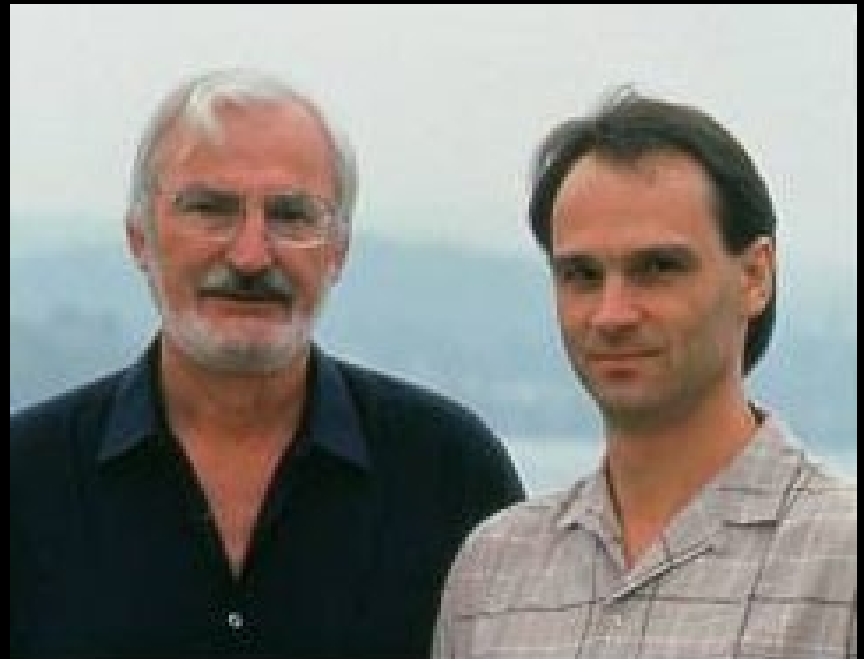
Laboratoire de Physique de la
Matière Vivante, EPFL

Département de Biologie Cellulaire et de
Morphologie, UNIL

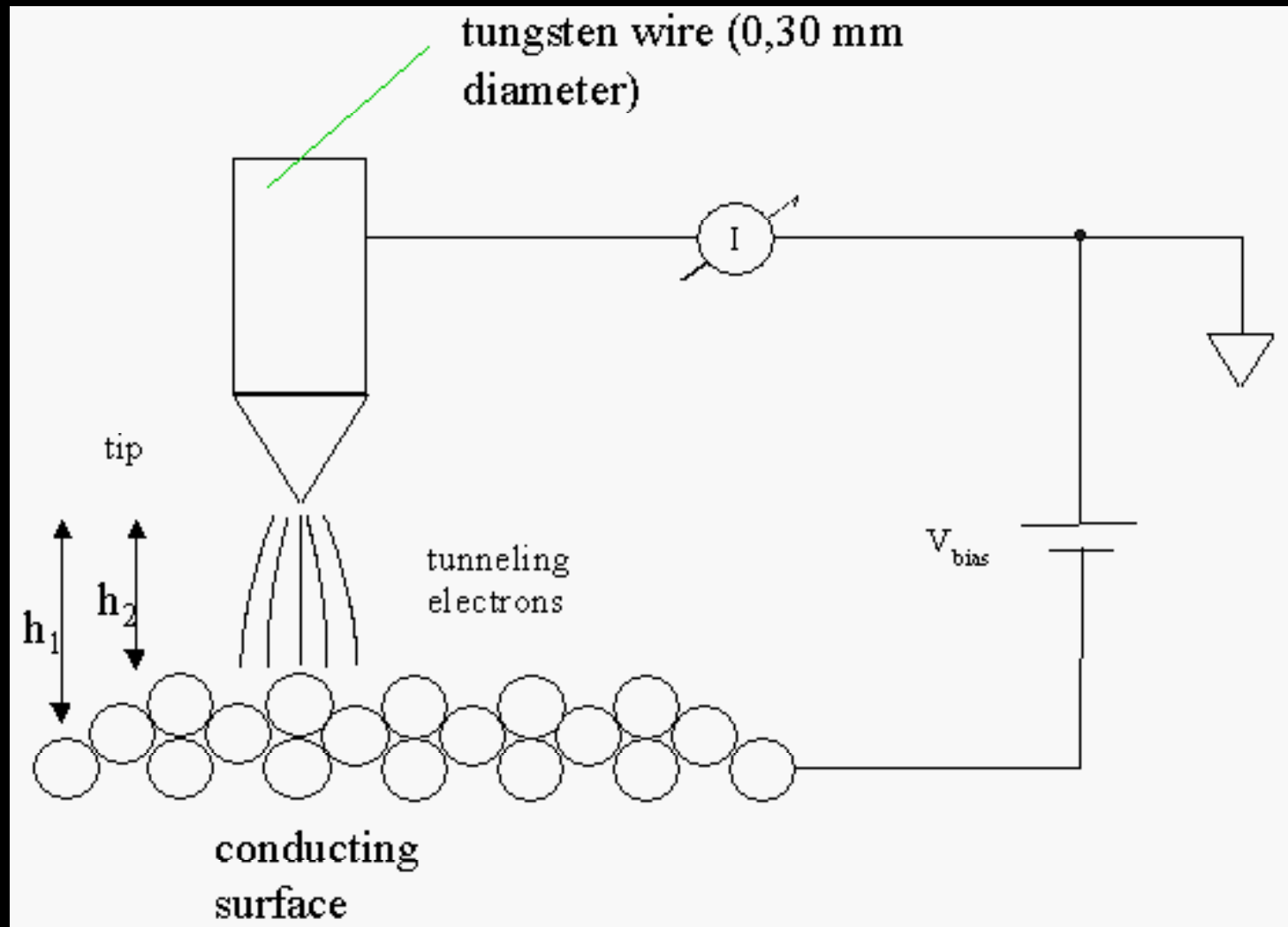
Microscopie à effet tunnel STM

1980 Binnig et Rohrer obtiennent les
premières images à résolution atomique

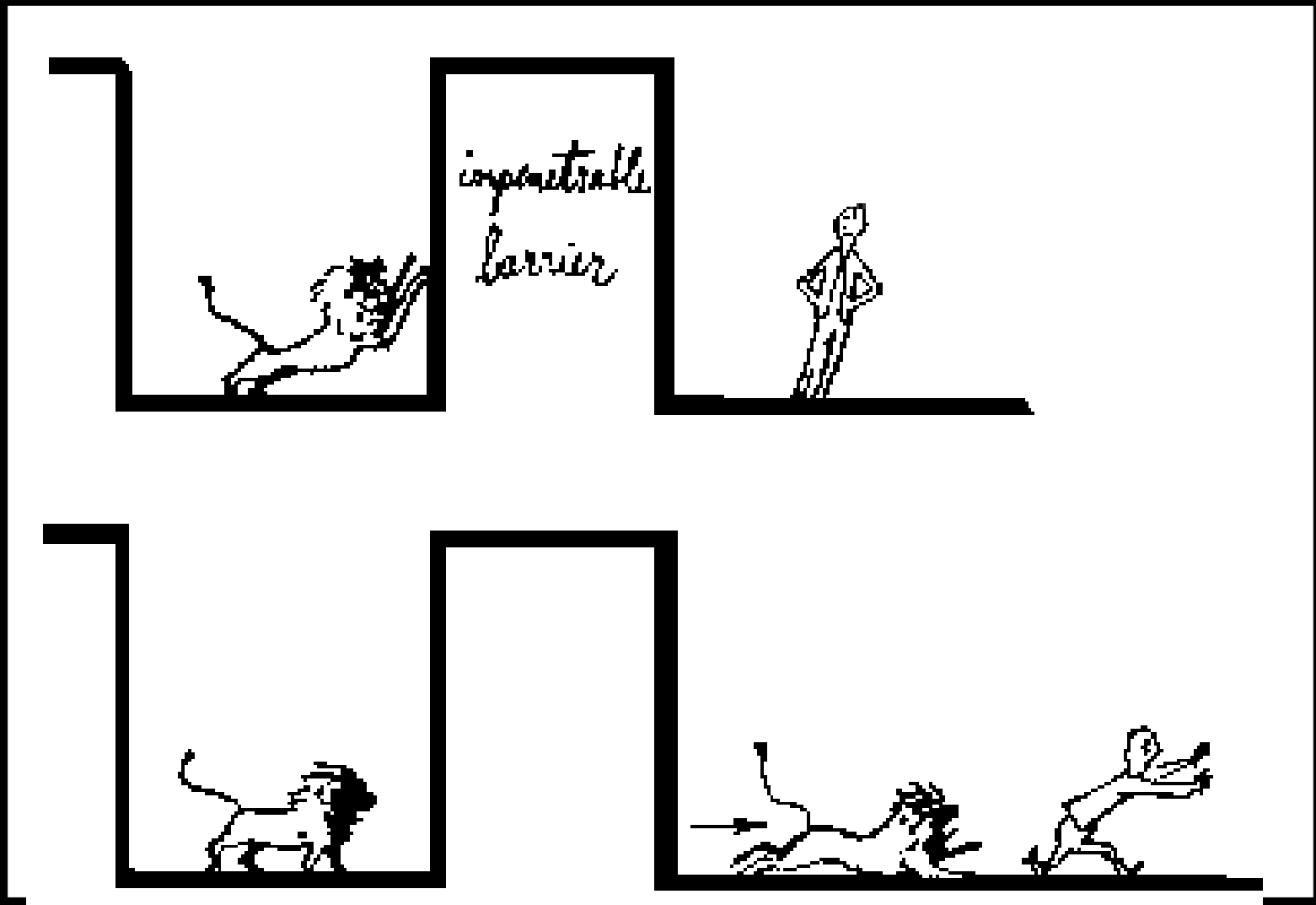
1986 Prix Nobel



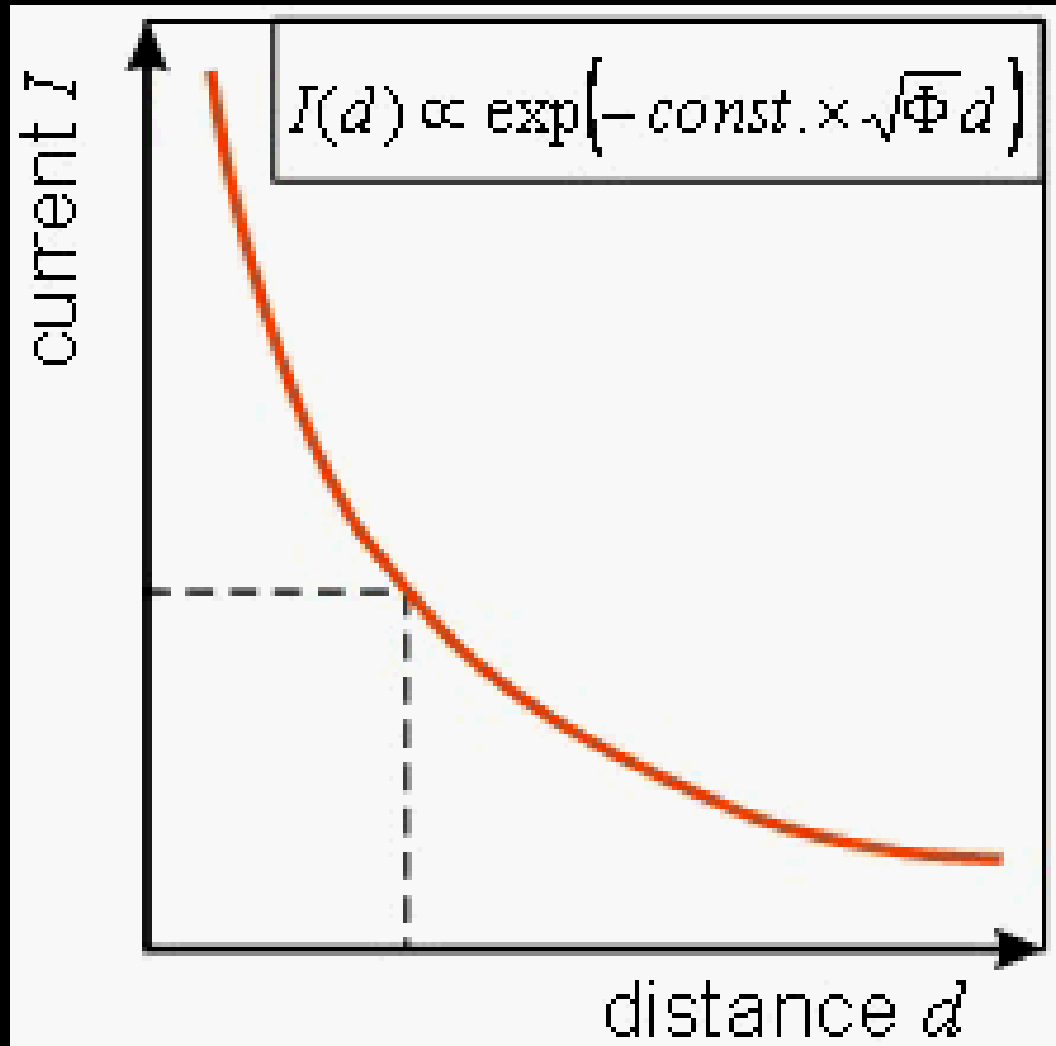
Principe de fonctionnement du STM



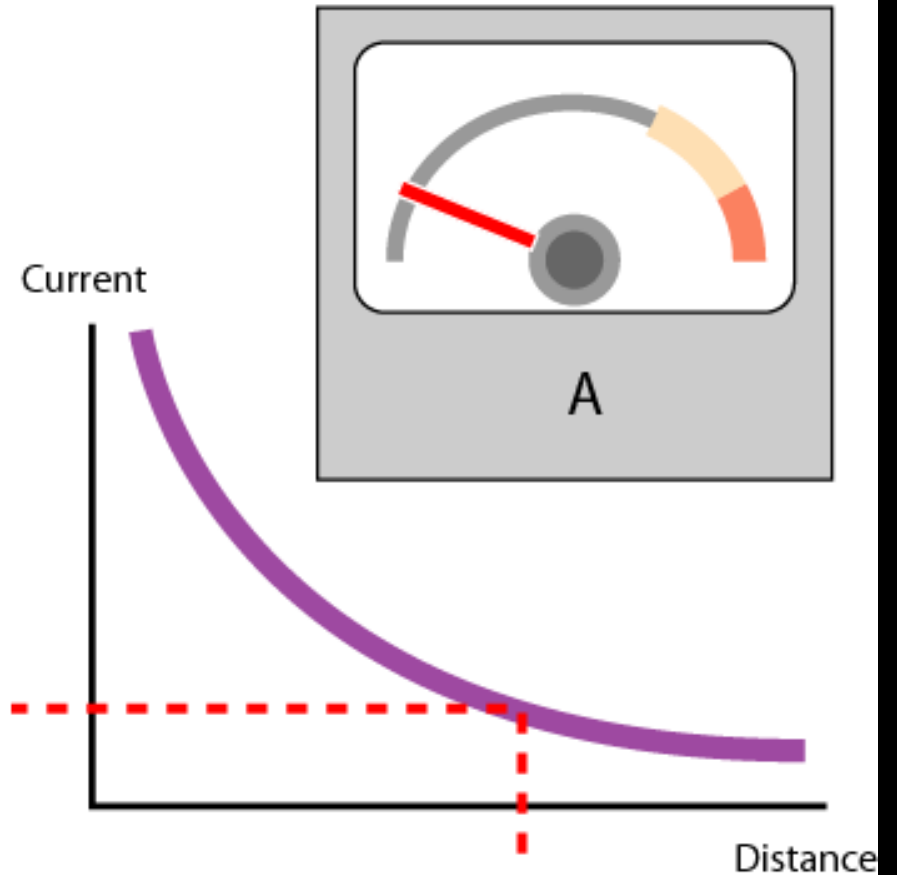
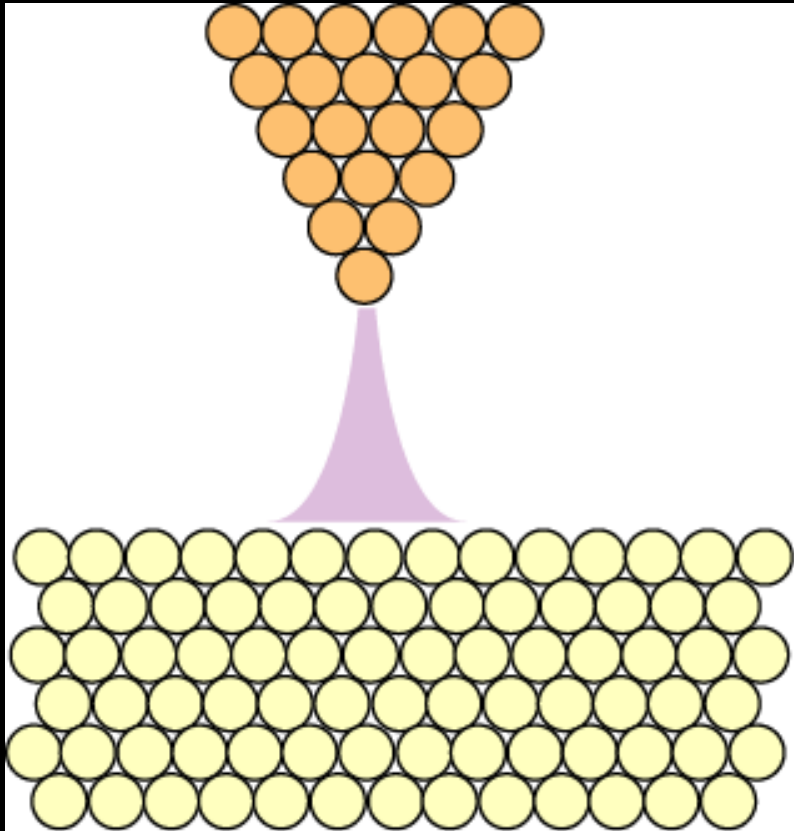
Différence entre la théorie classique et quantique



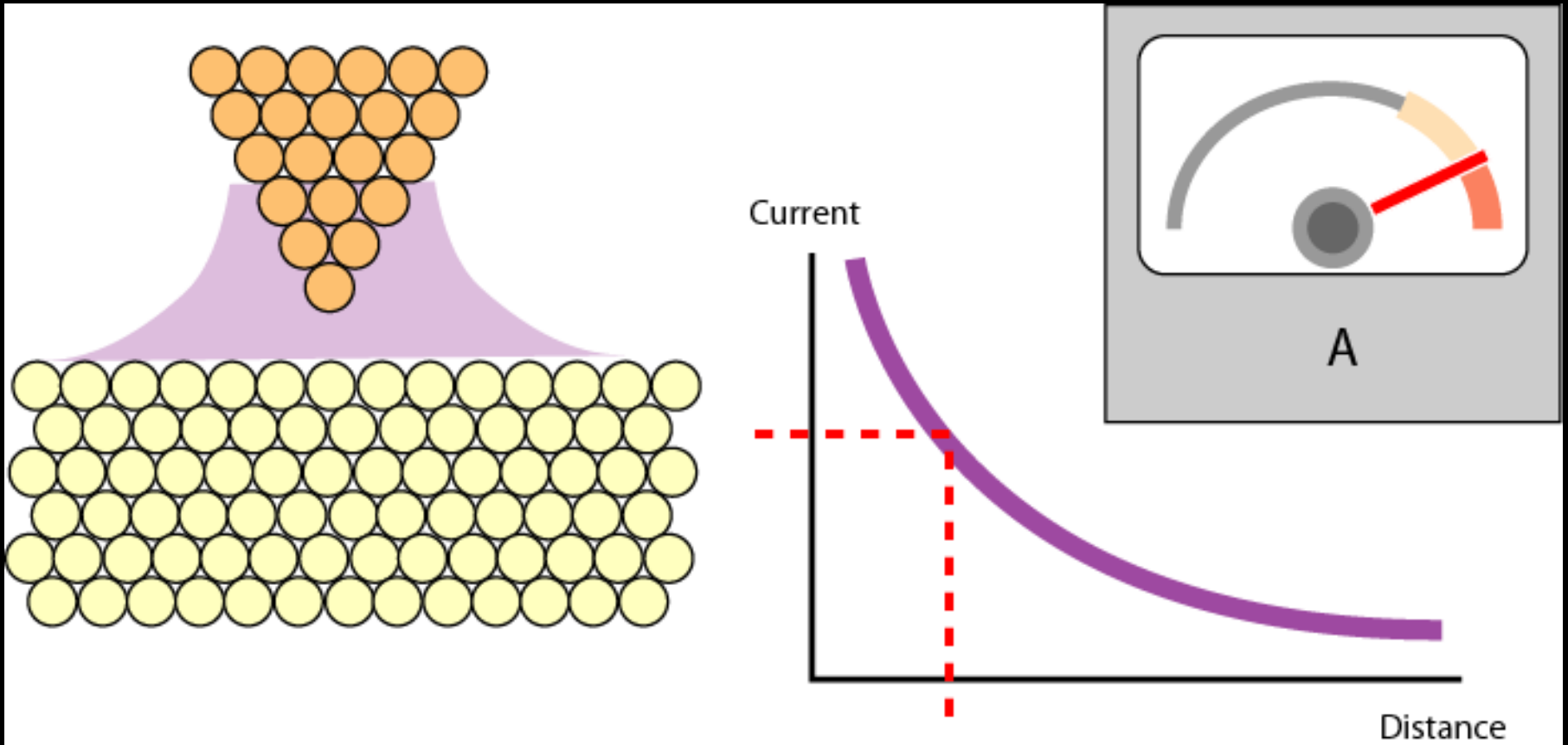
Dépendence du courant tunnel en fonction de la distance



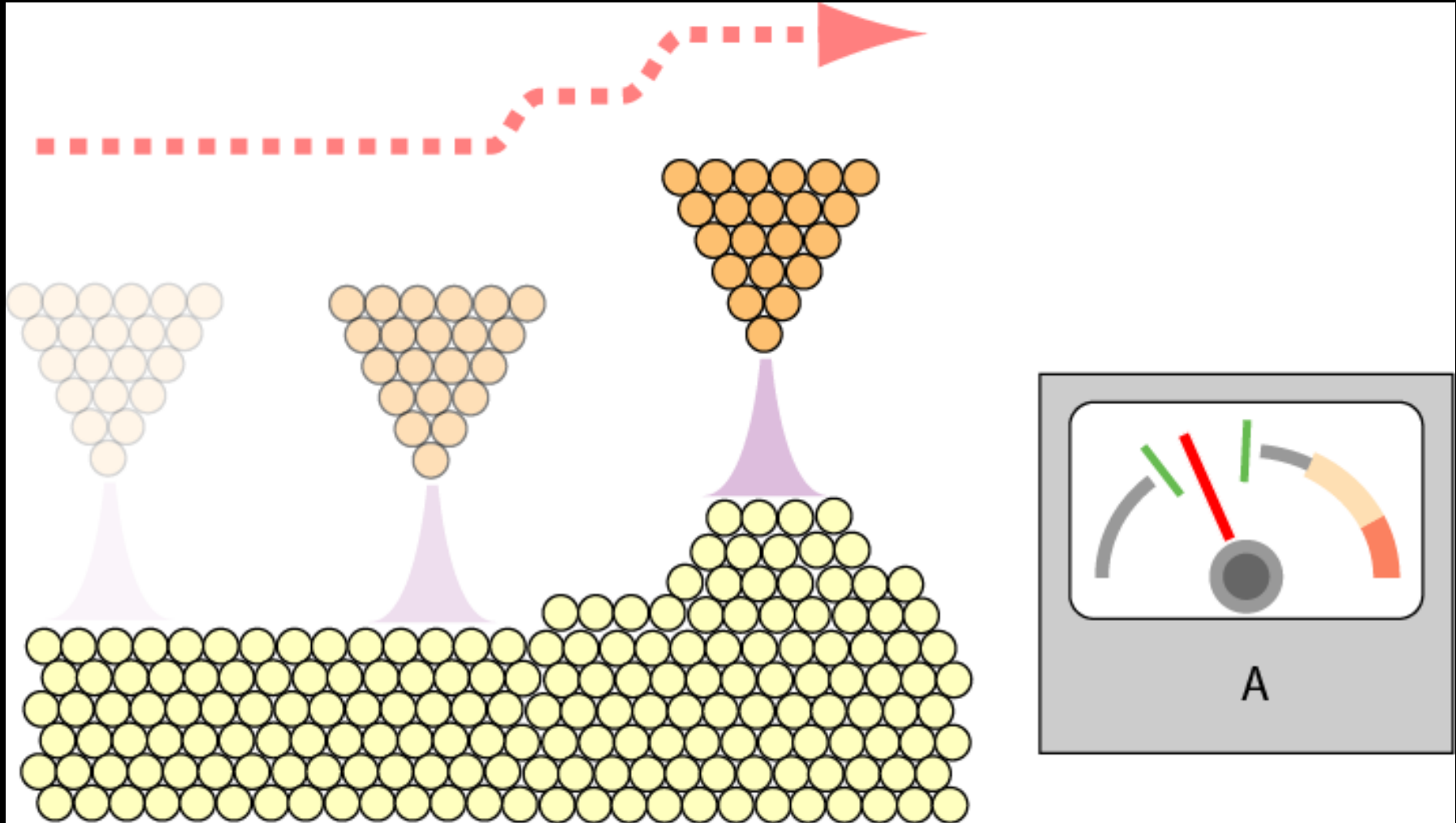
Current = f (Distance)



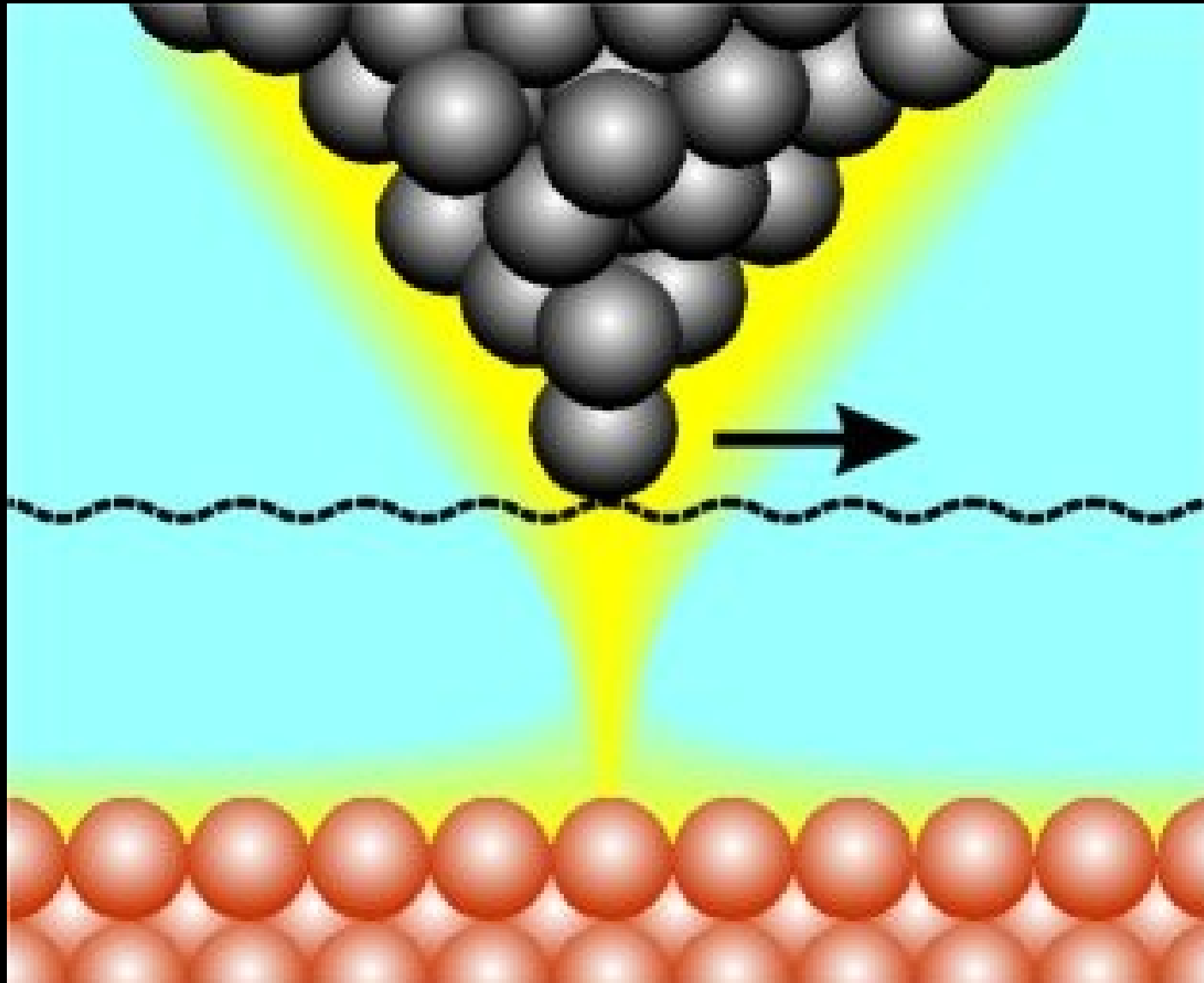
Current = f (Distance)



Scanning

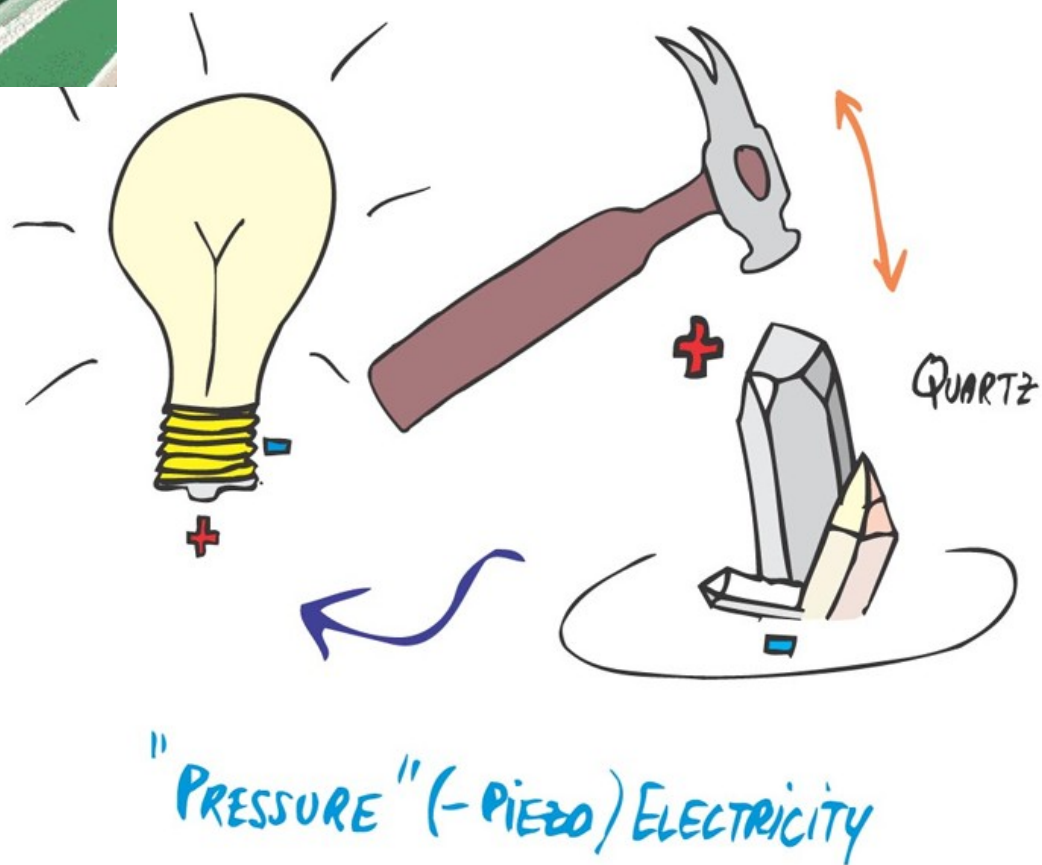


STM





Piezo



Principe de fonctionnement des cristaux piezzoélectriques

Conversion tension - déplacement

Conversion déplacement - tension

Scanner

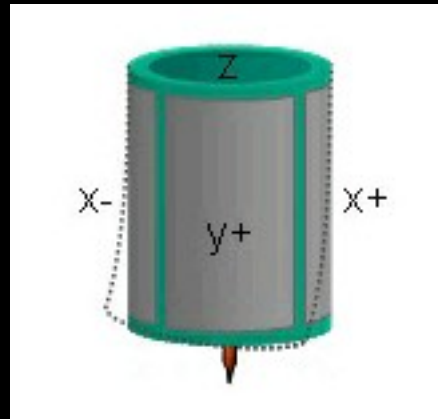
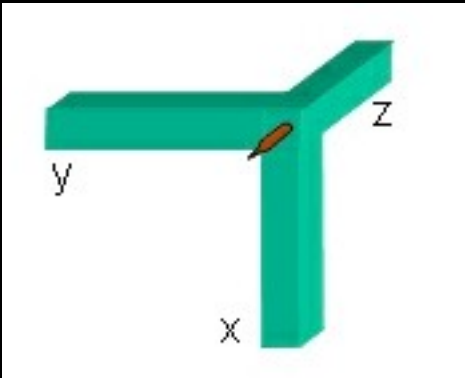
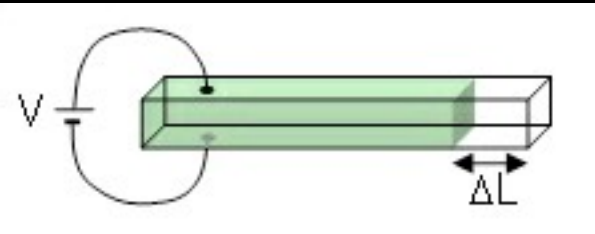
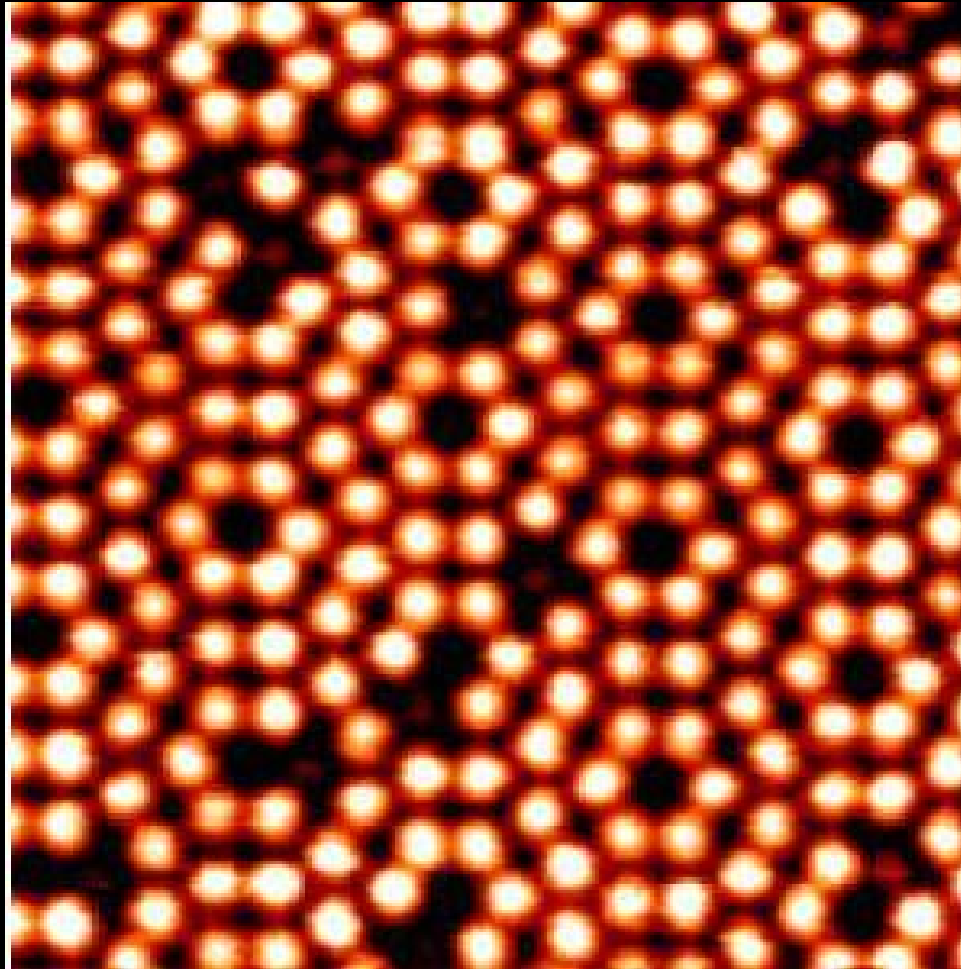
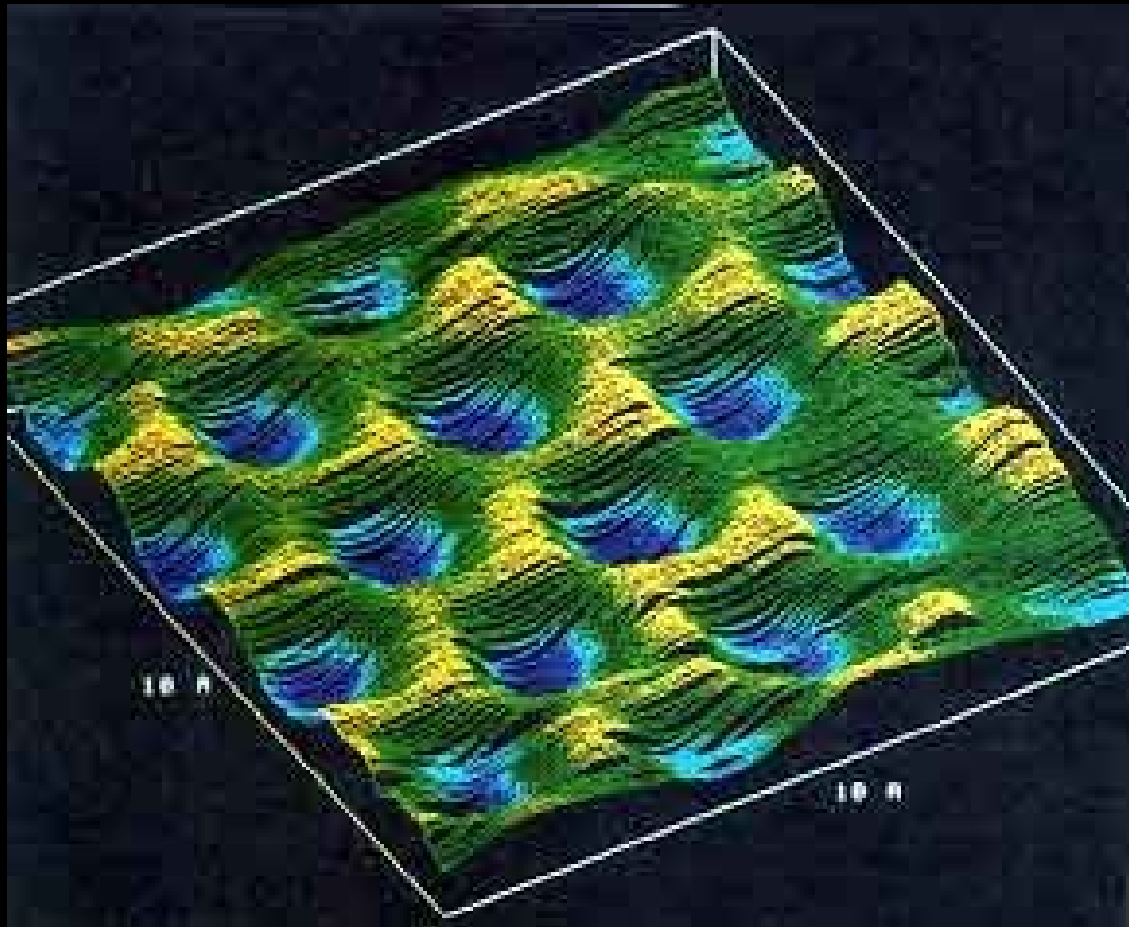


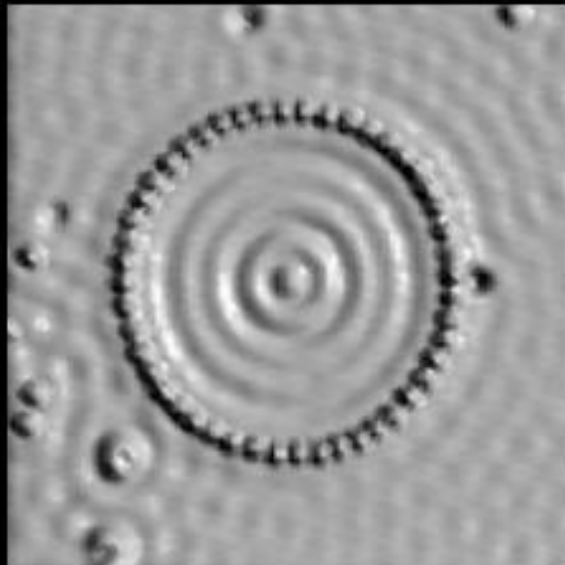
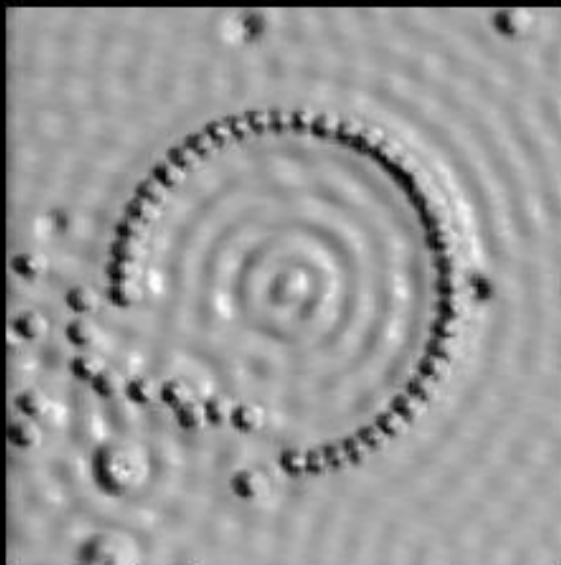
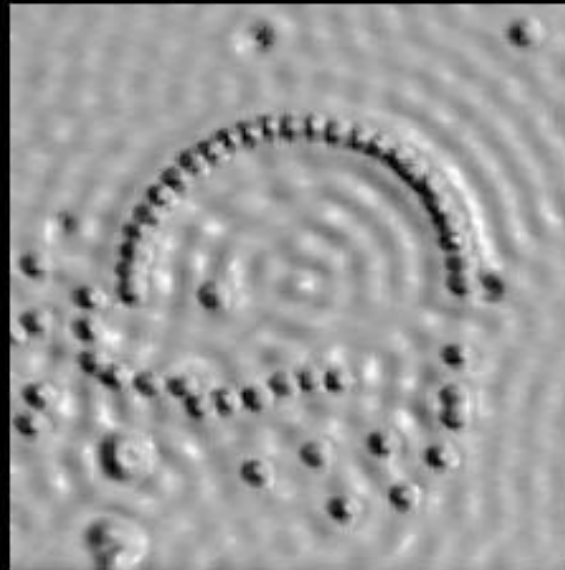
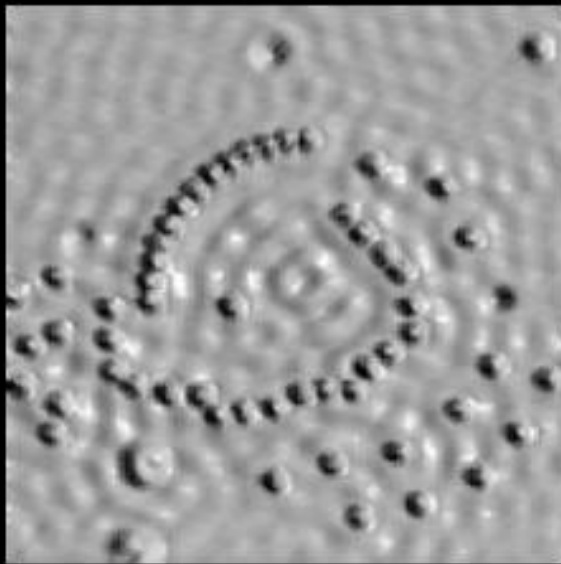
Image STM du Si(111)7x7



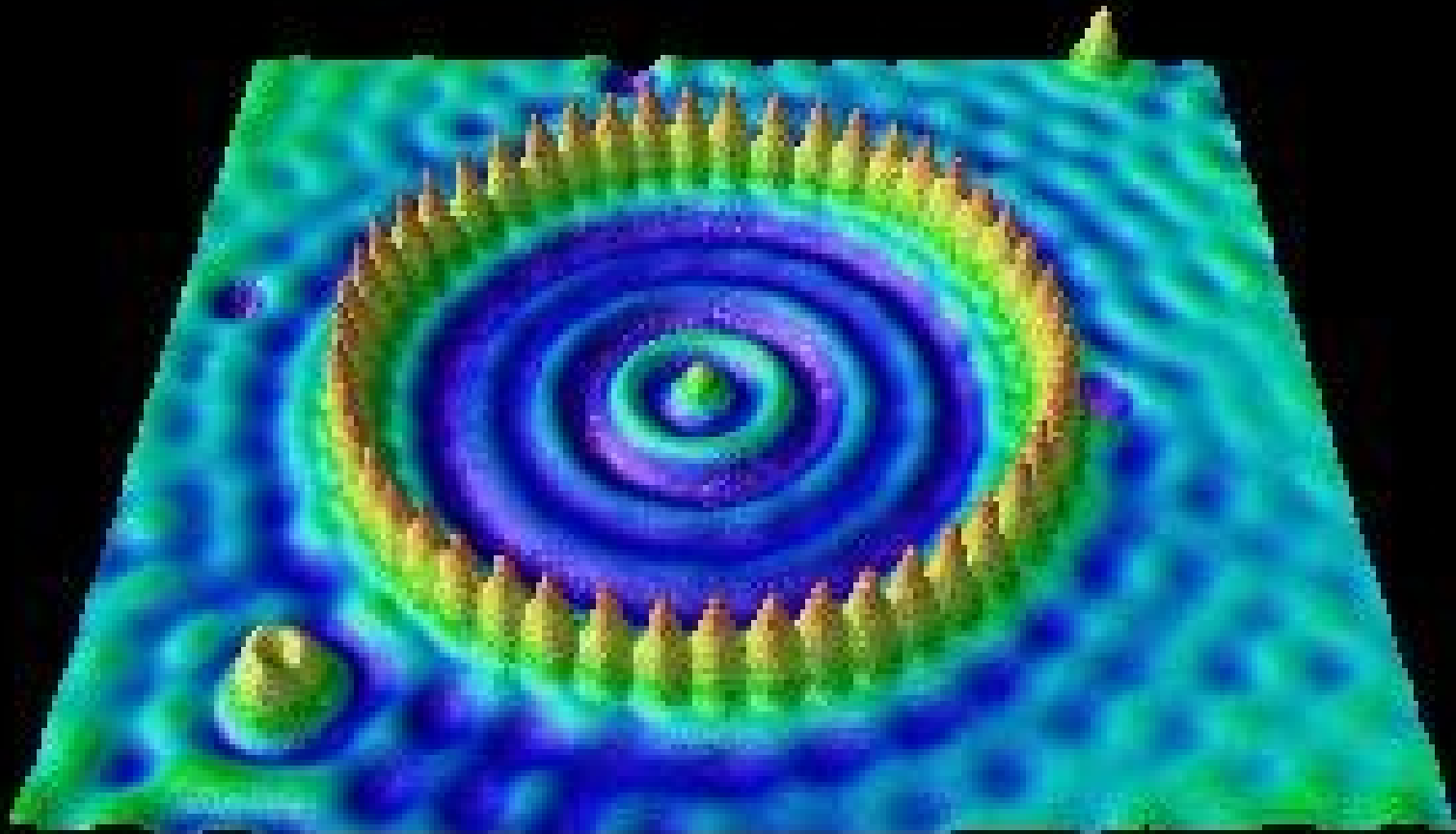
Graphite



The making of ...



STM



UHV STM



Avantages

résolution spatiale élevée

possibilité de manipuler des atomes

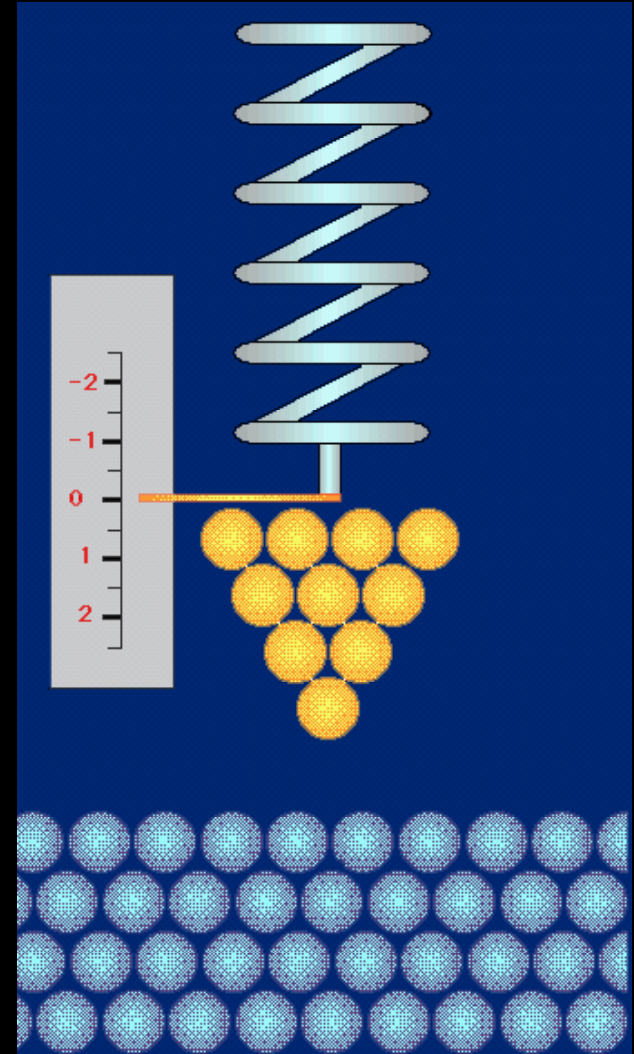
Désavantages

marche pas sur des échantillons isolants

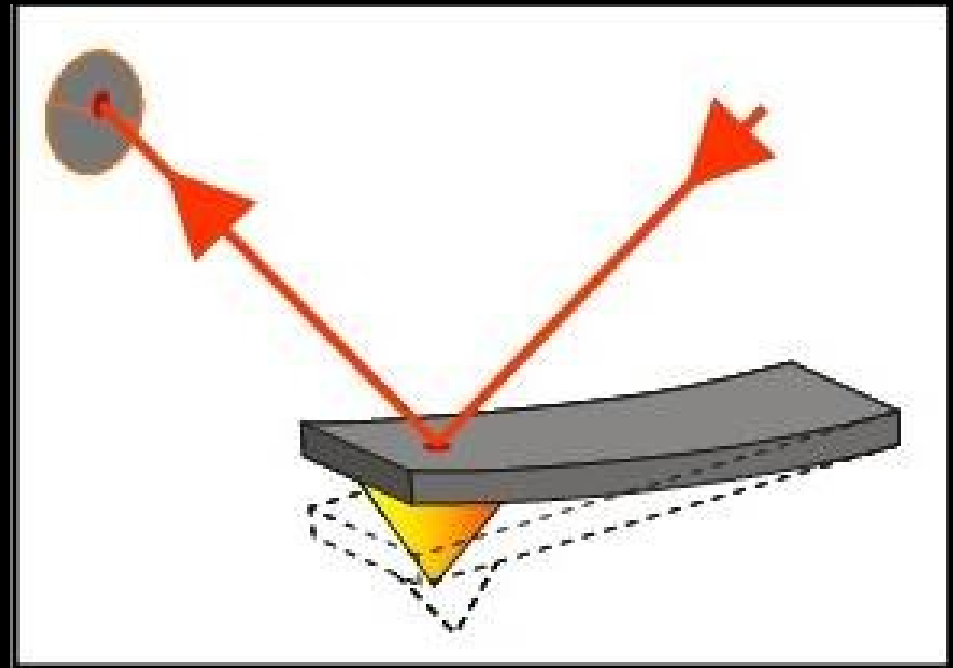
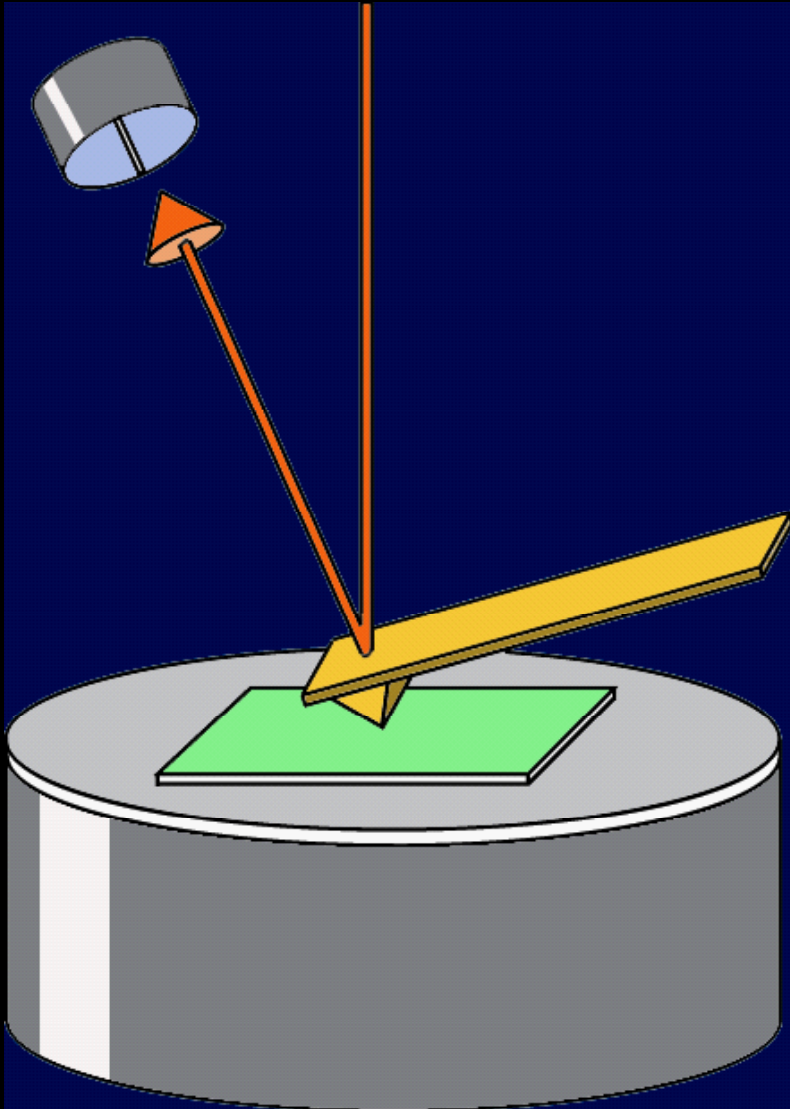
marche pas dans l'eau

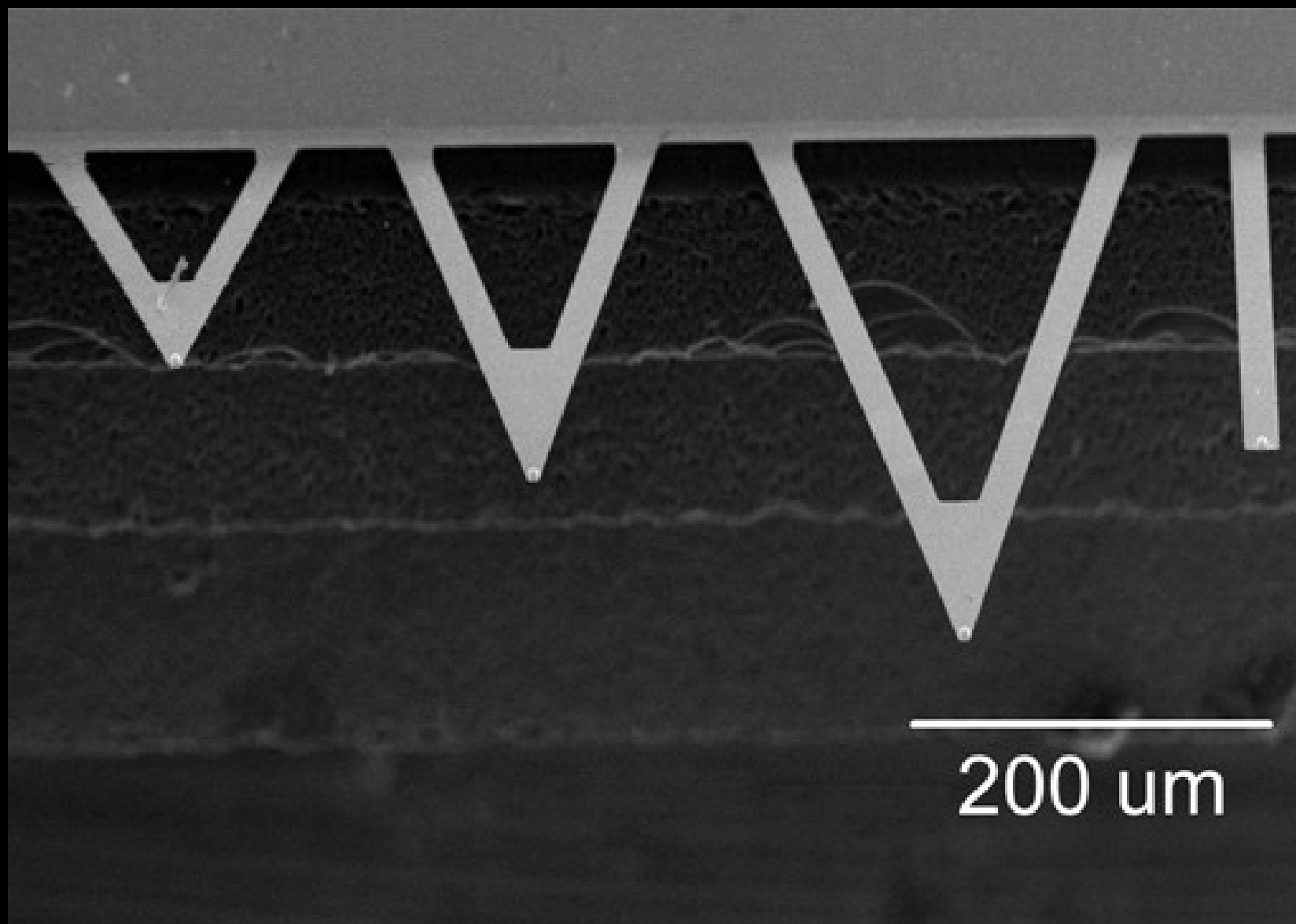
Principe de fonctionnement de l'AFM

Senseur de forces d'interaction

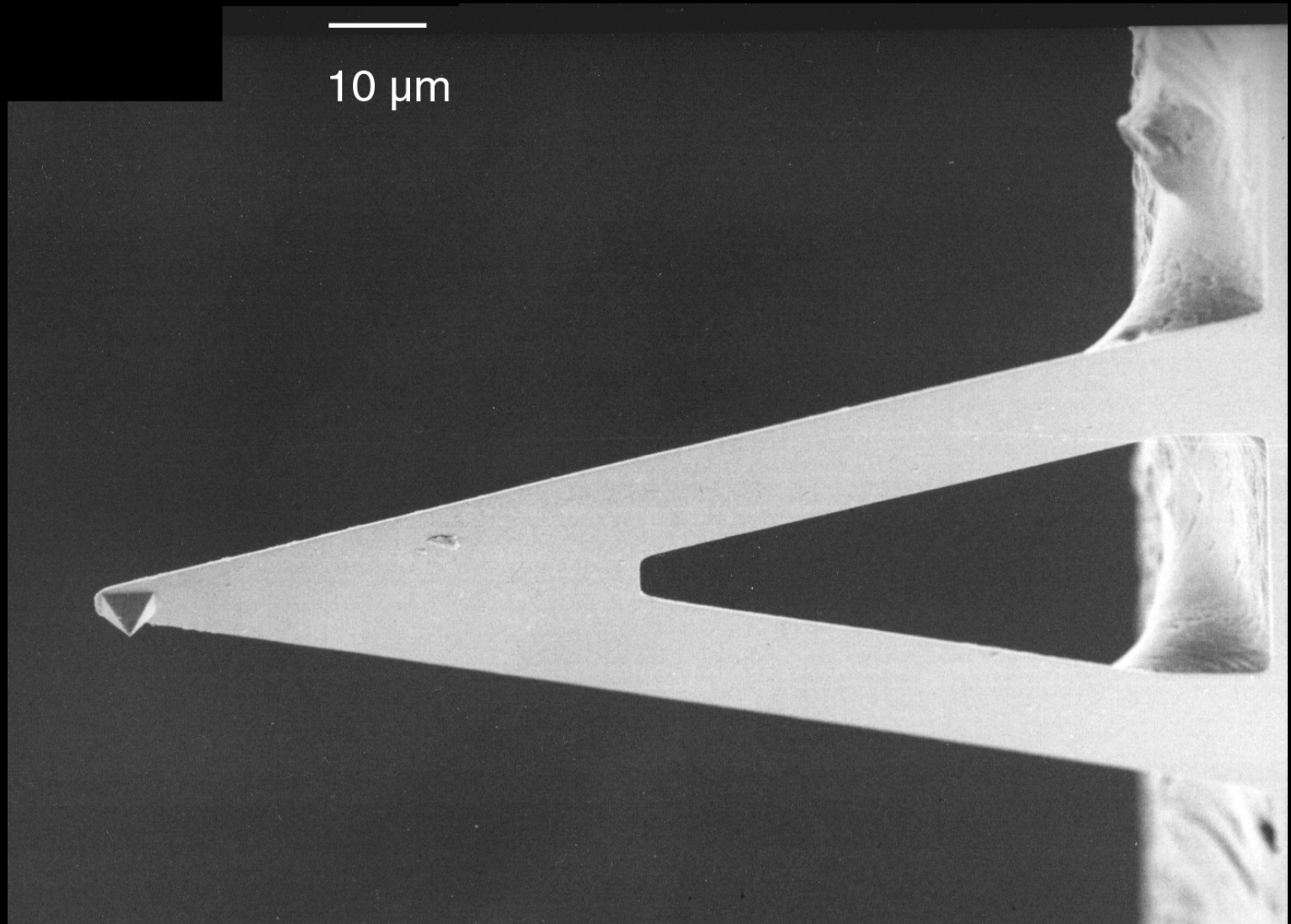


Implémentation

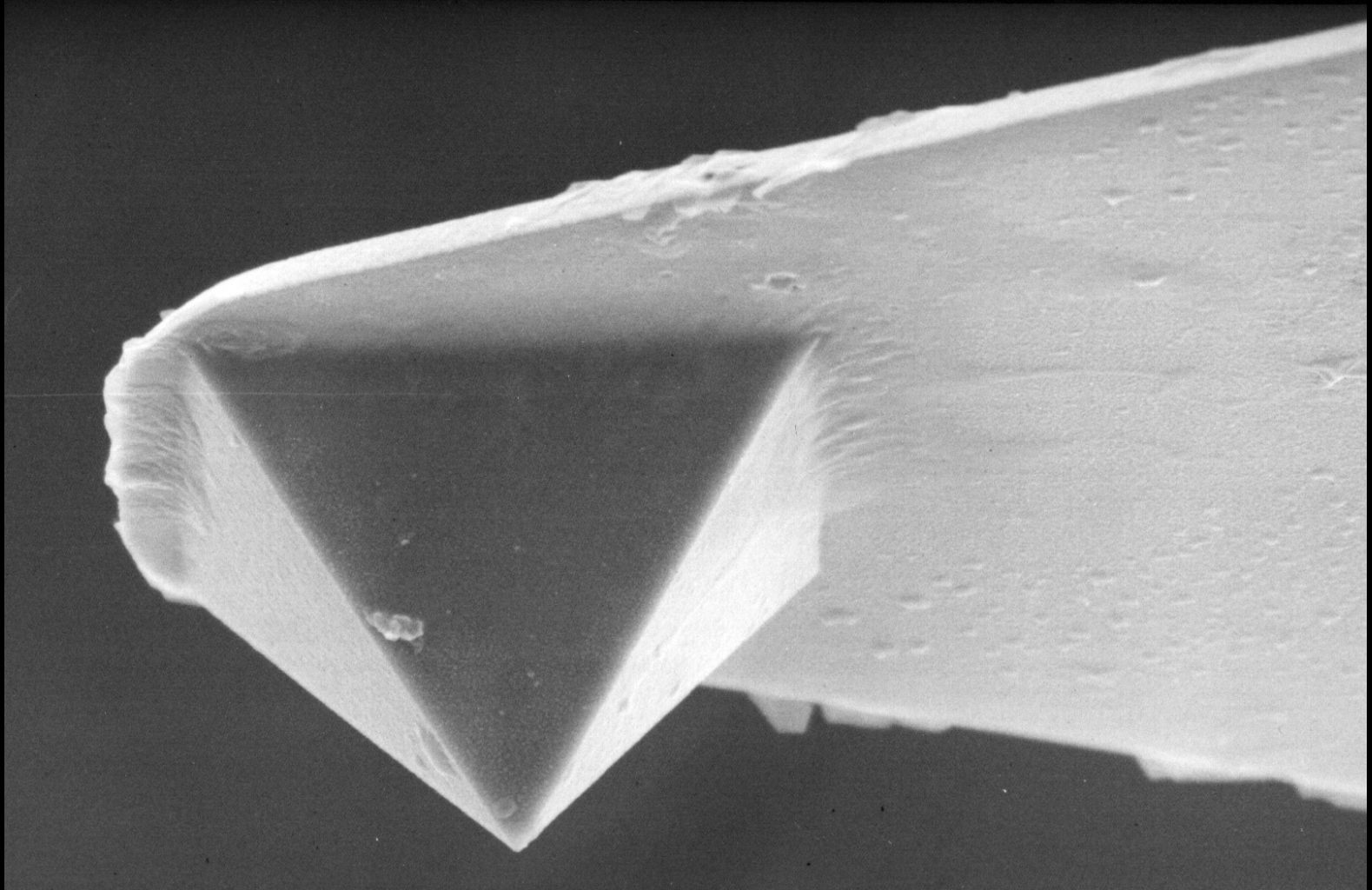




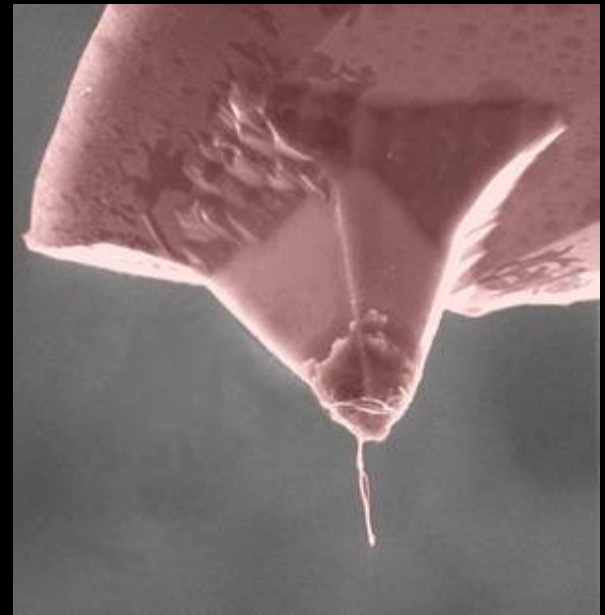
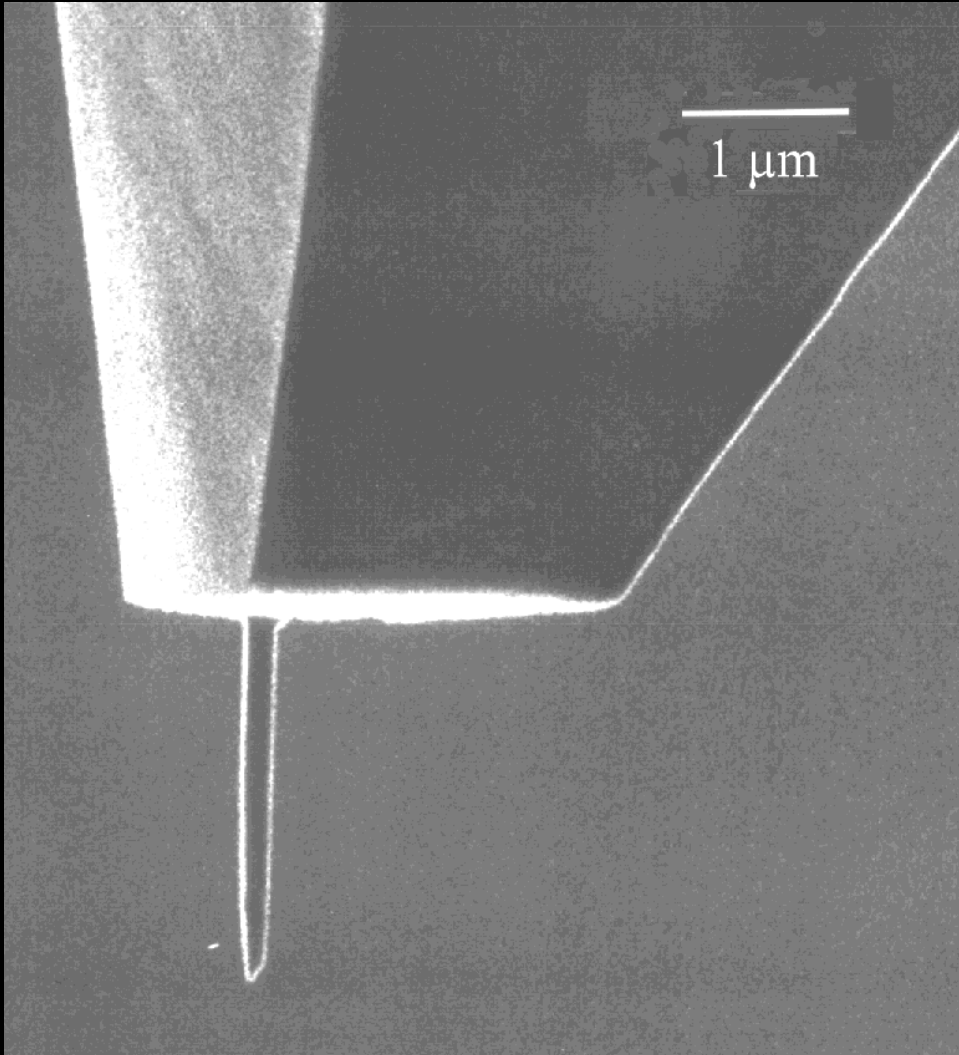
Le levier



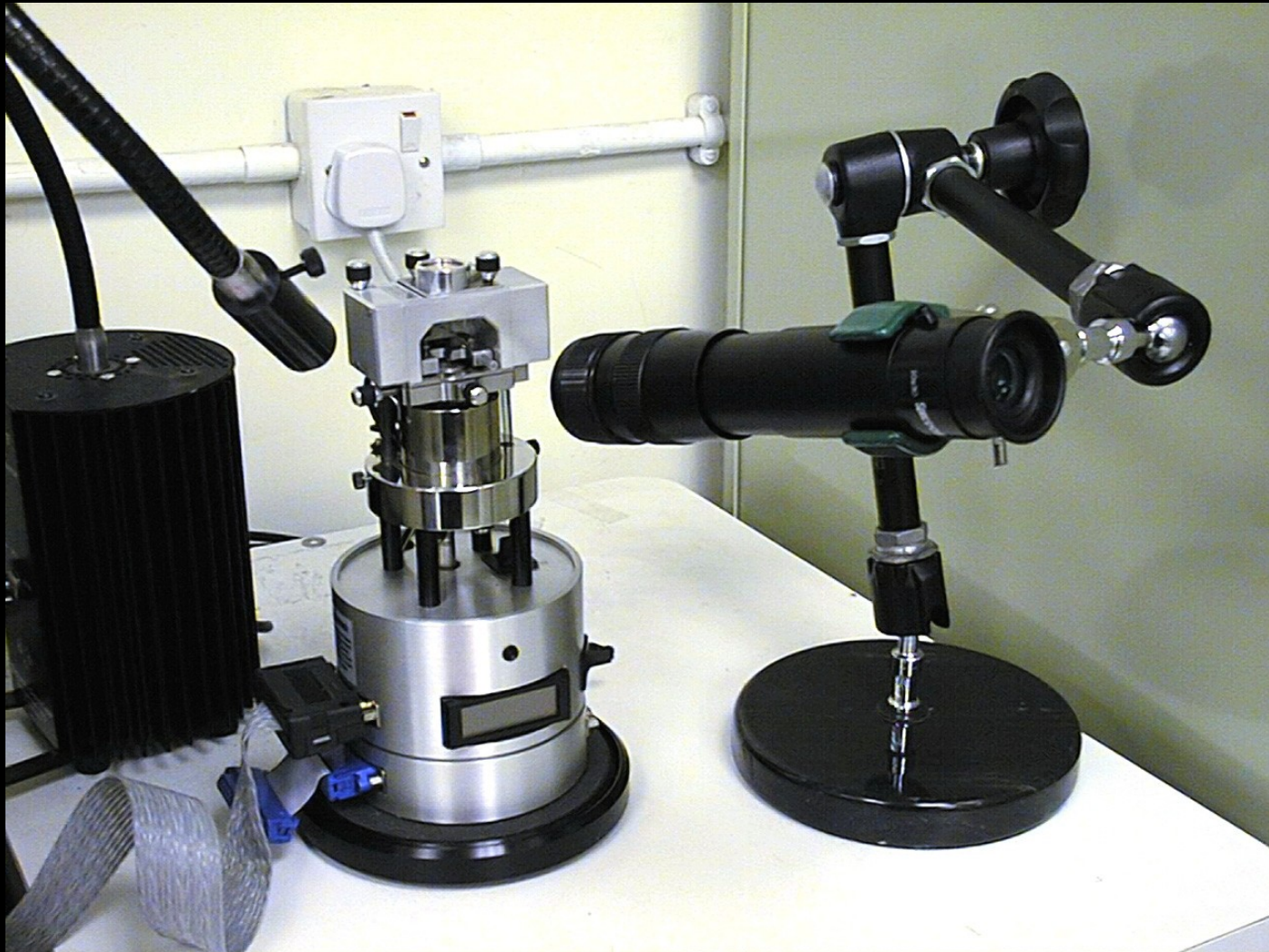
pointe

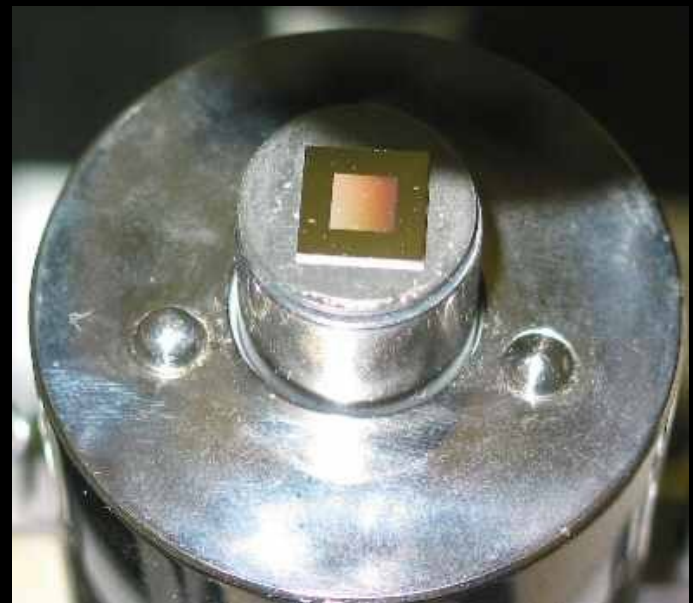


Super-pointe



Nanoscope

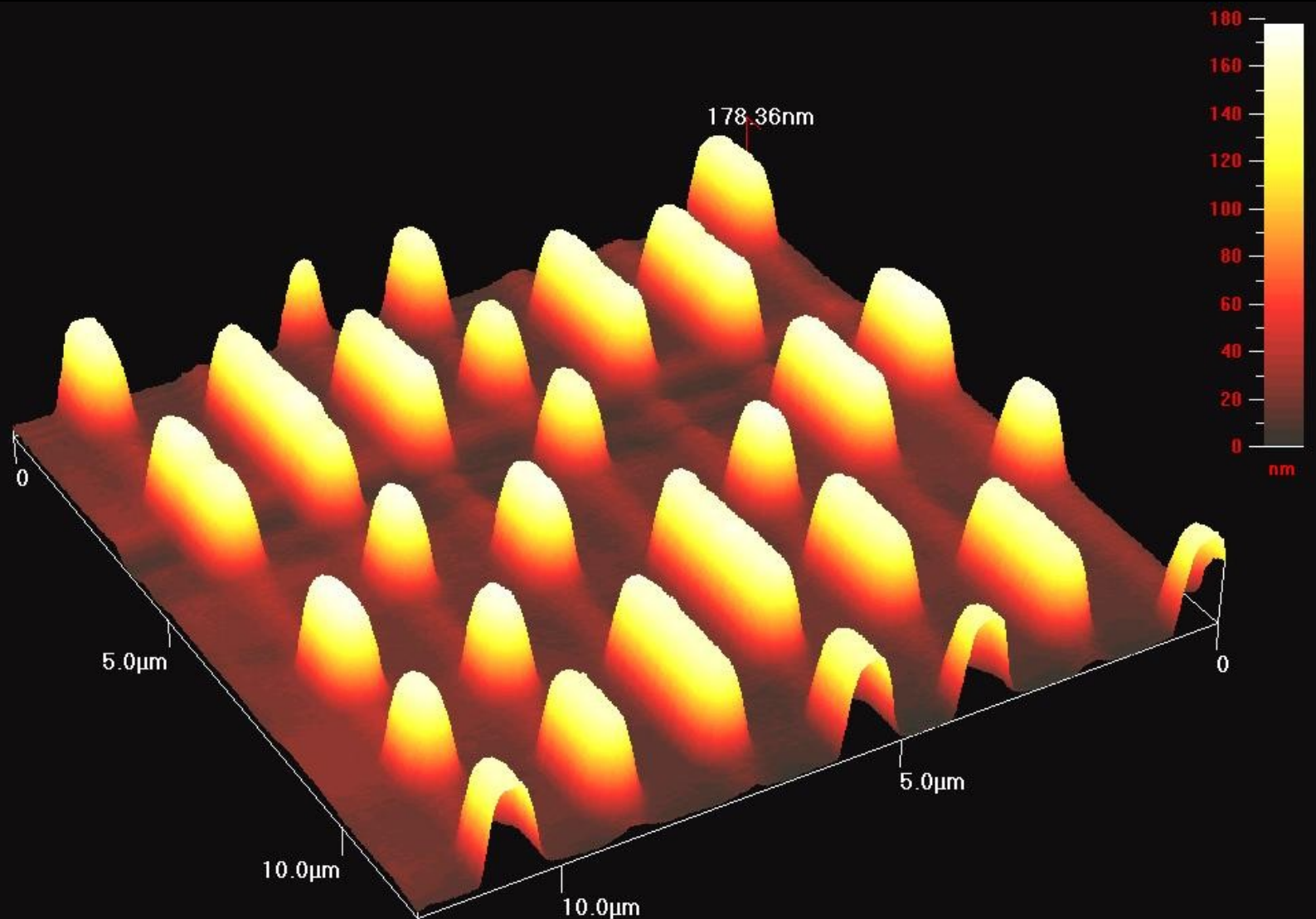




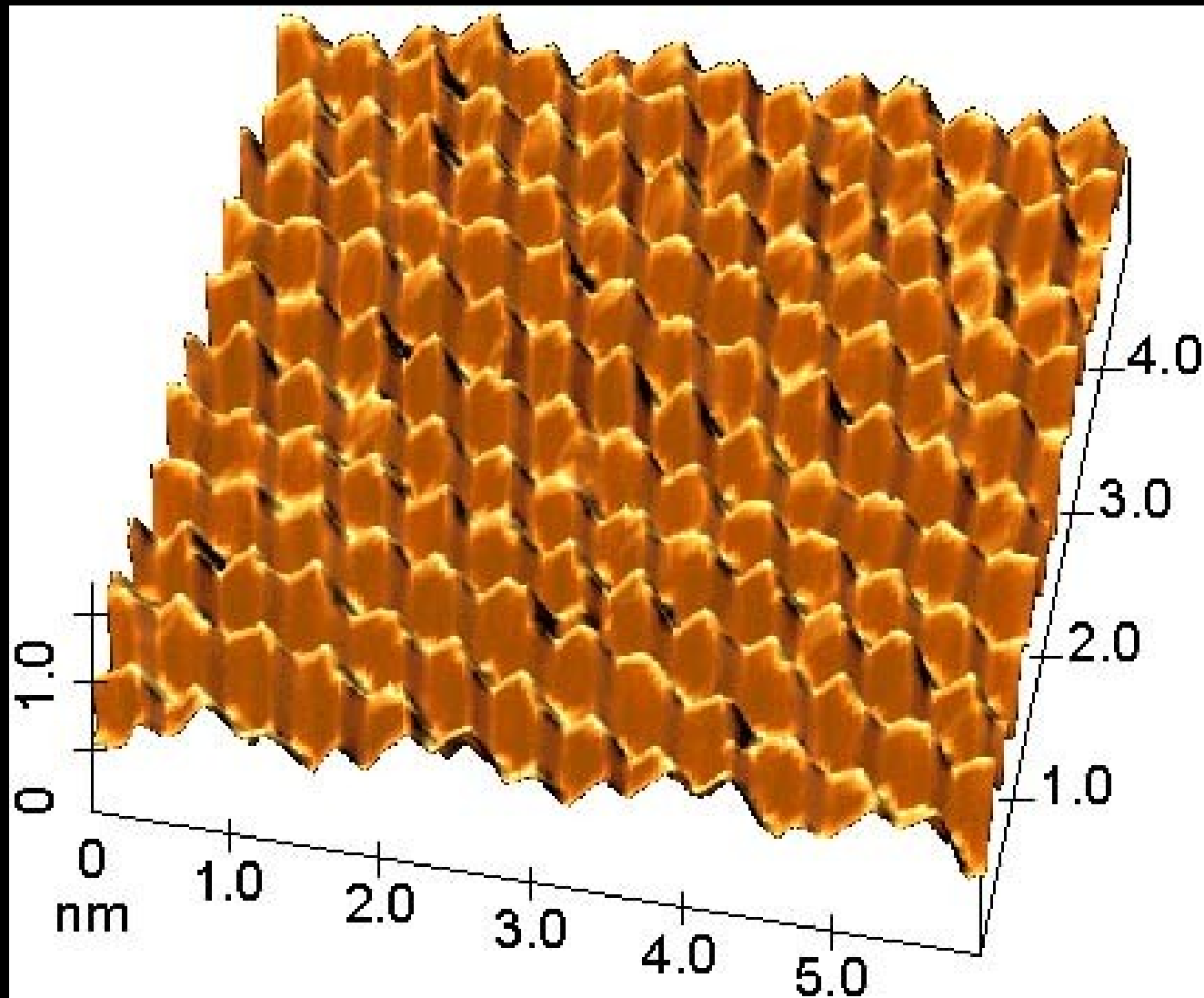
AFM



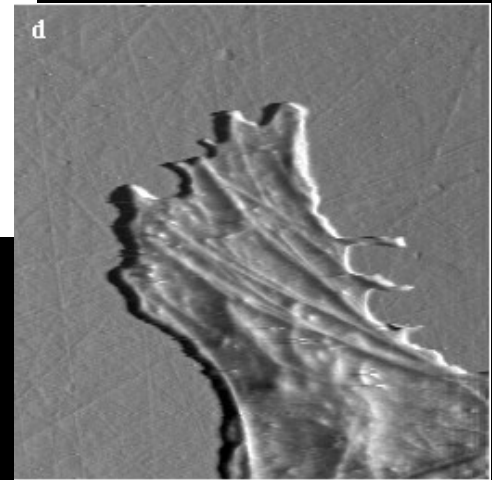
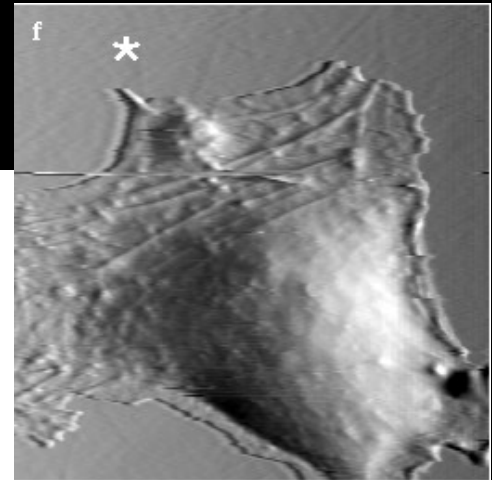
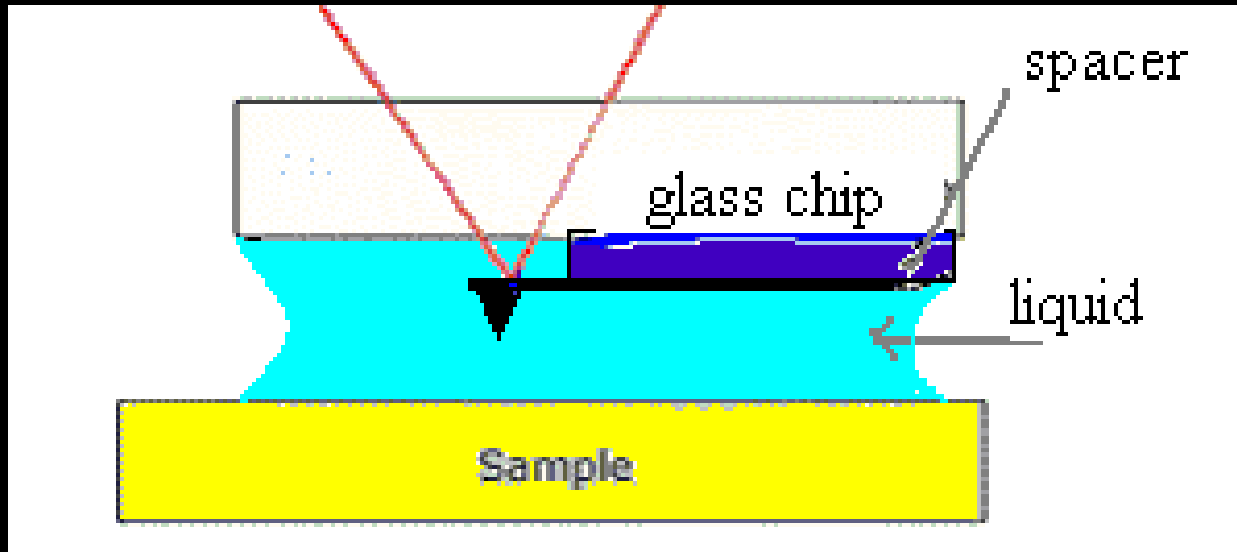
CD



Mica



Imagerie dans des liquides

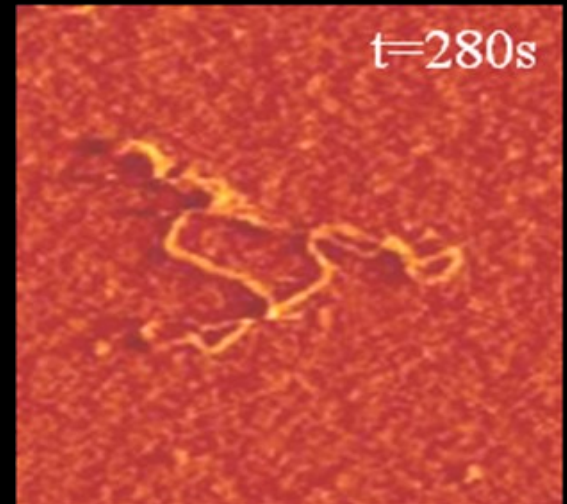
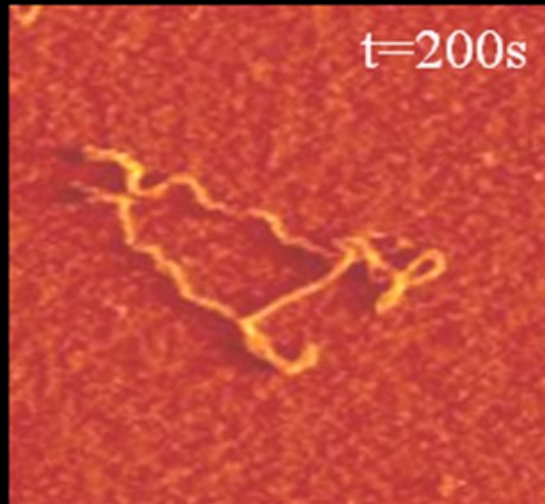
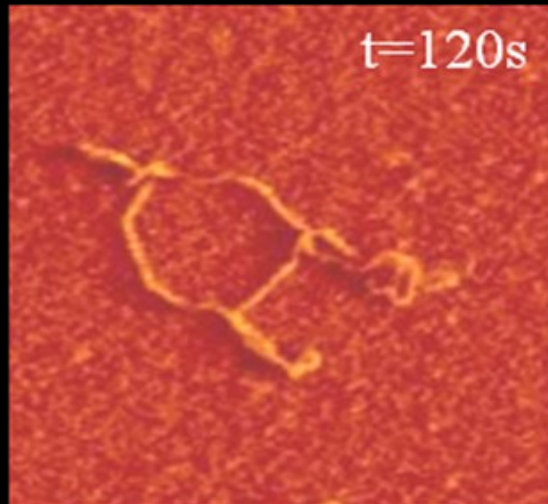
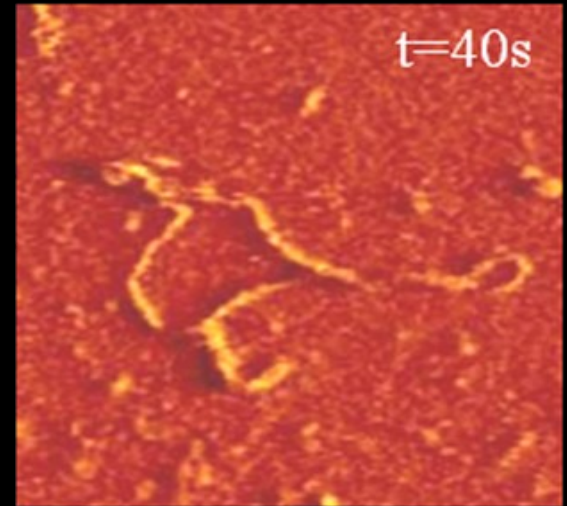
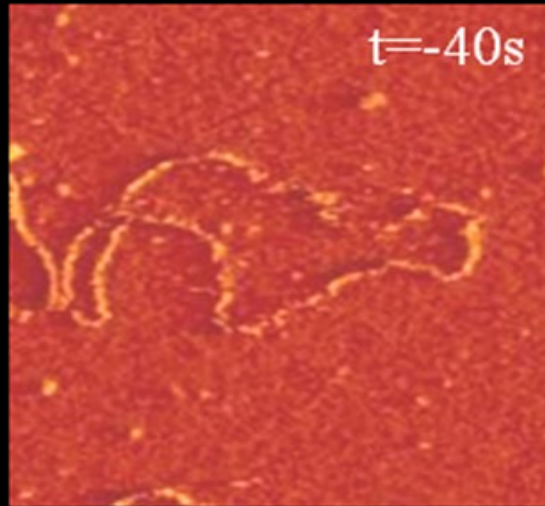
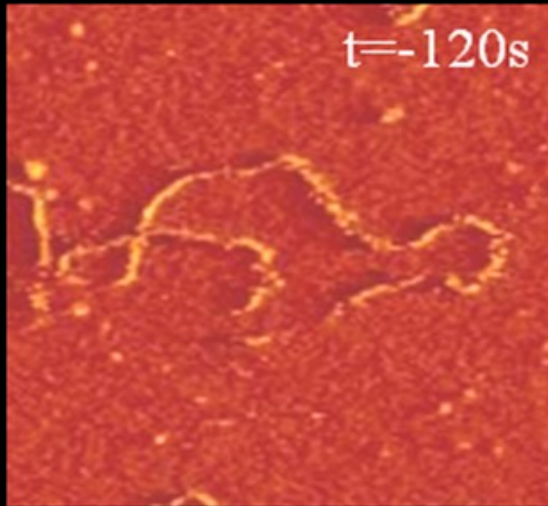


Injection system

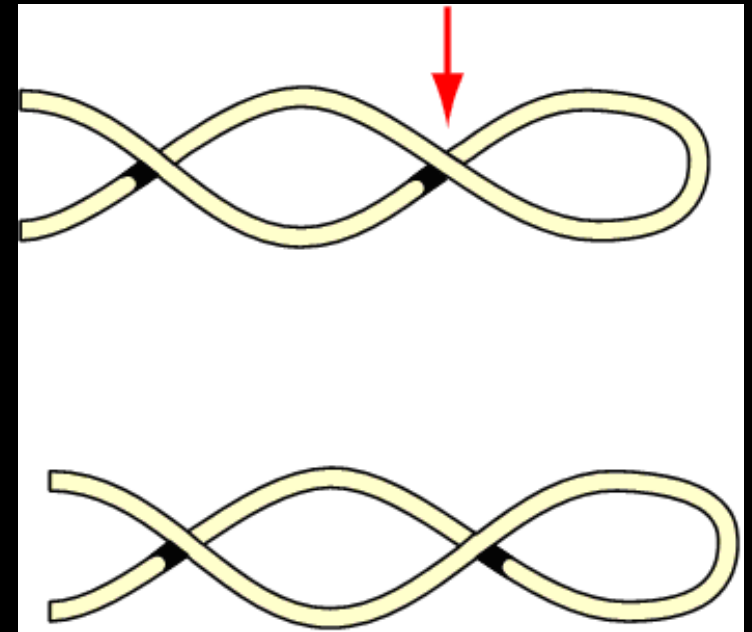
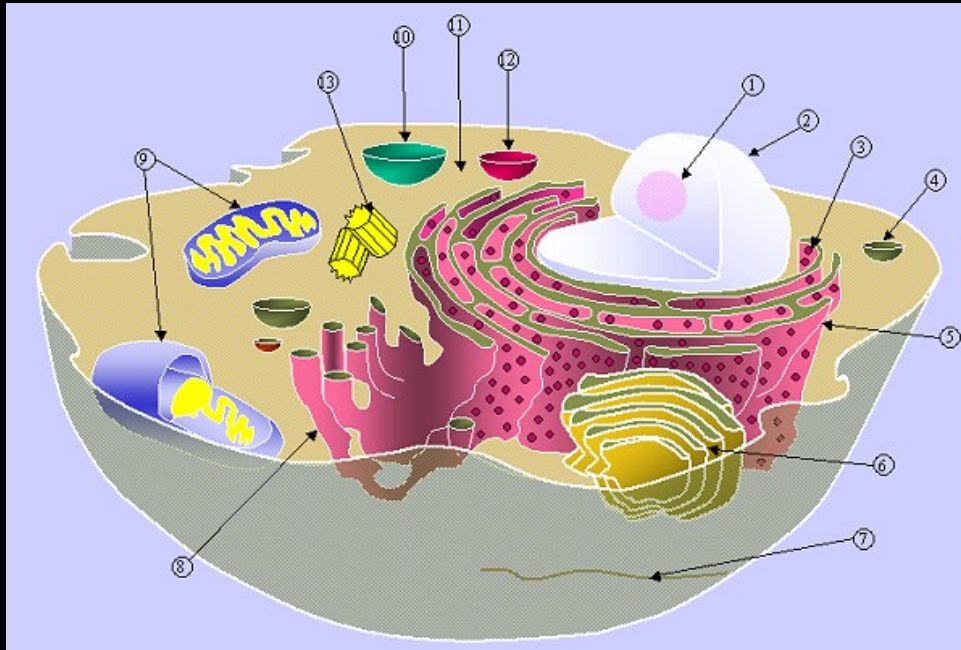
C. Haeberli



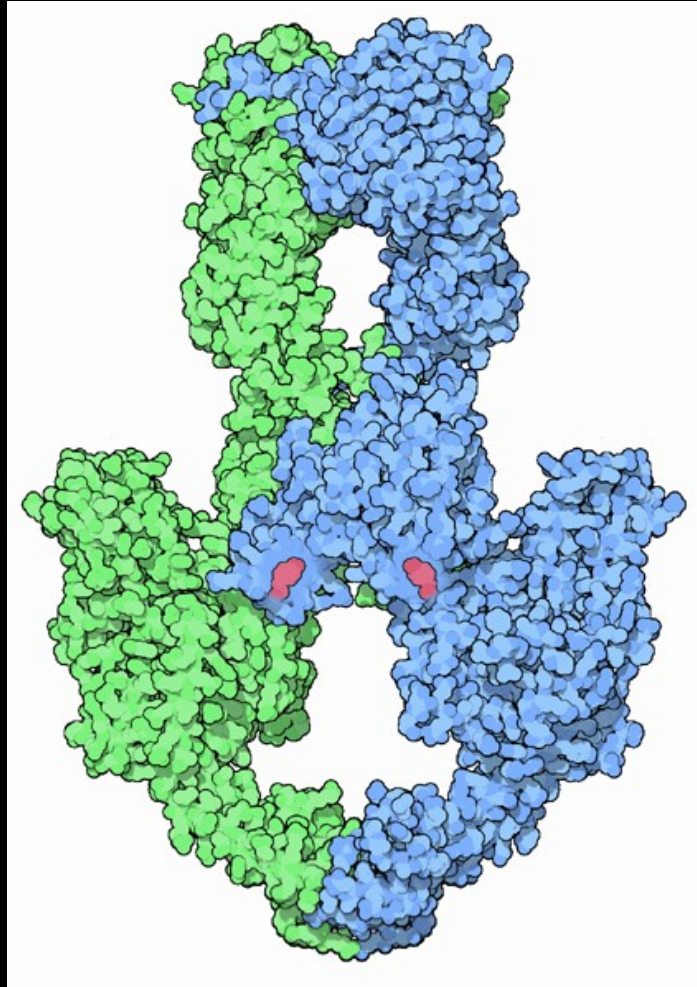
Effect de la Daunorubicine sur l'ADN



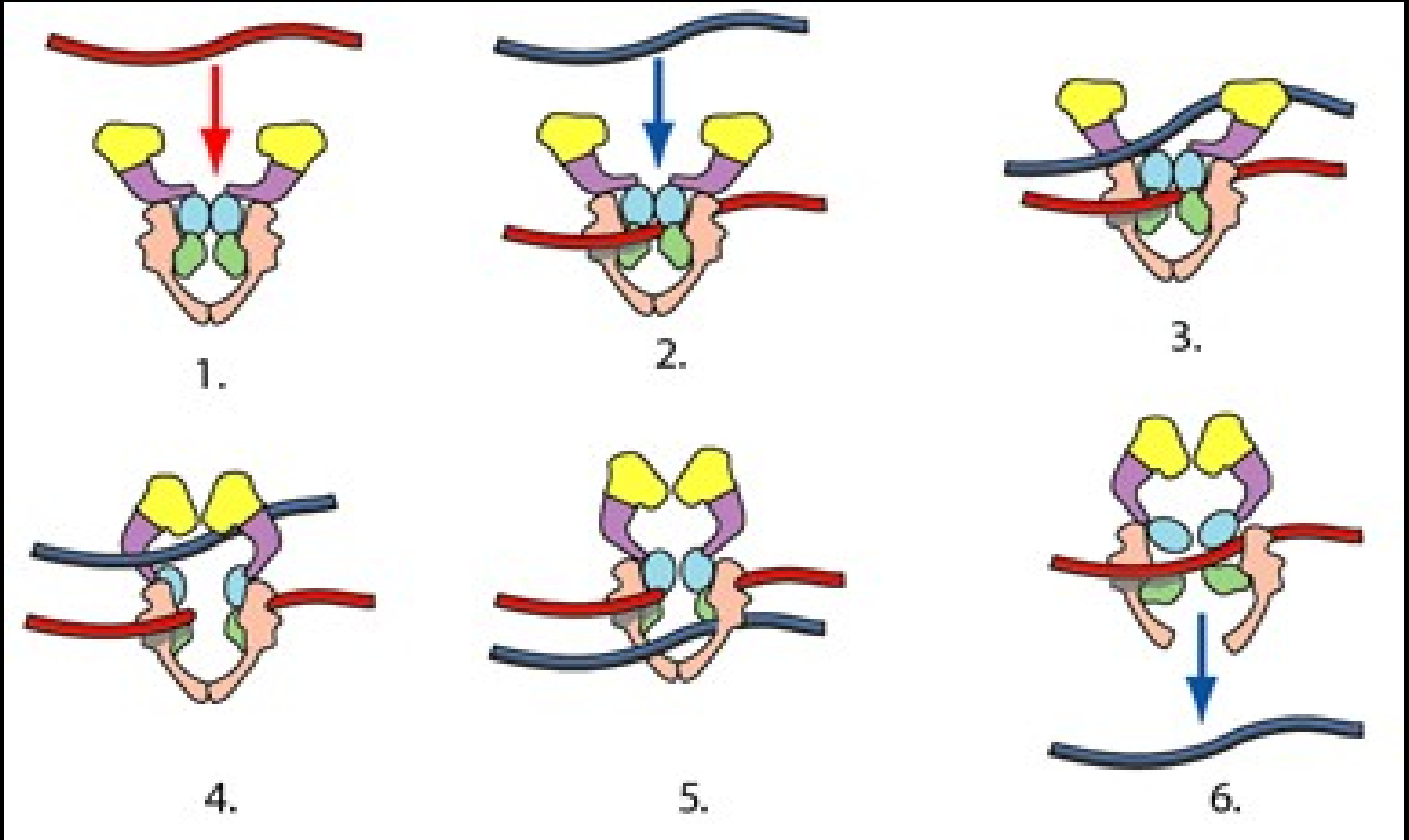
Fonction de la Topoisomerase II



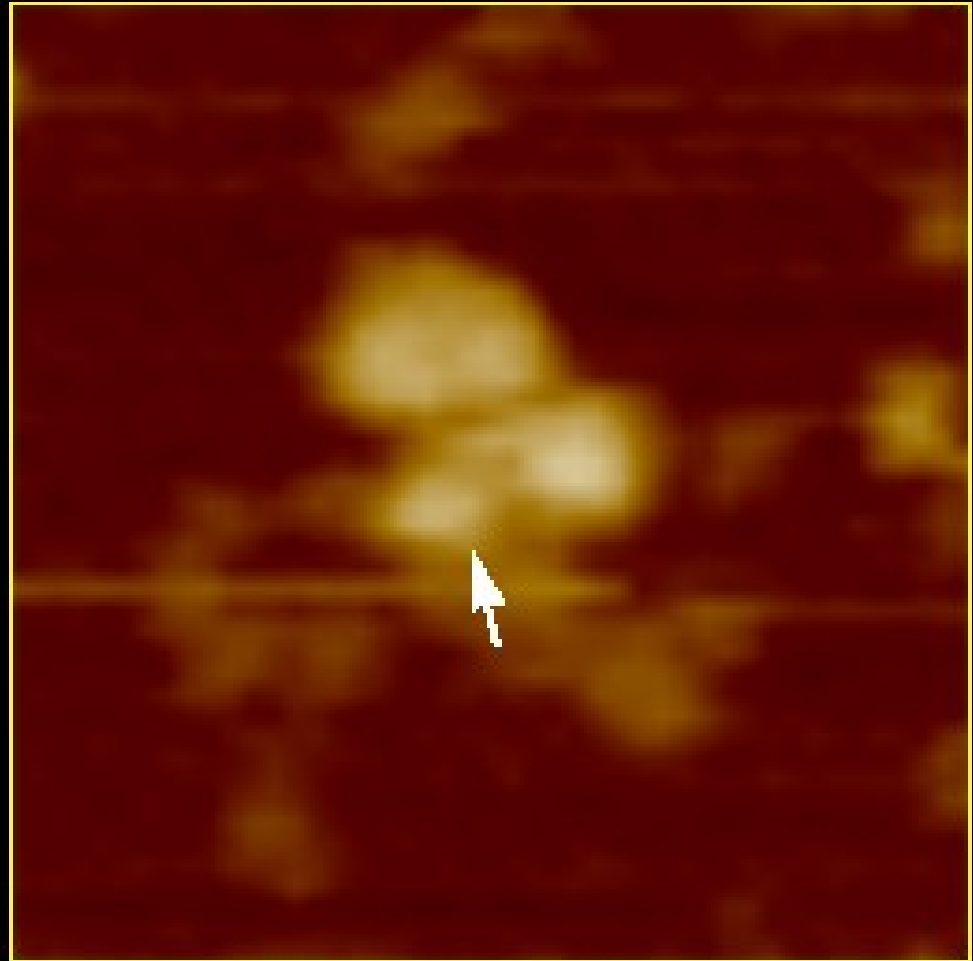
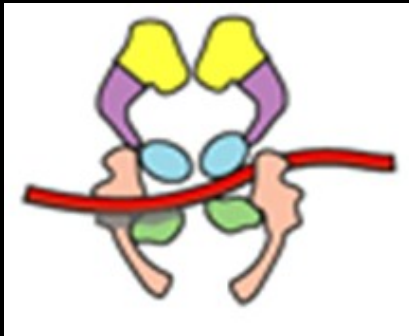
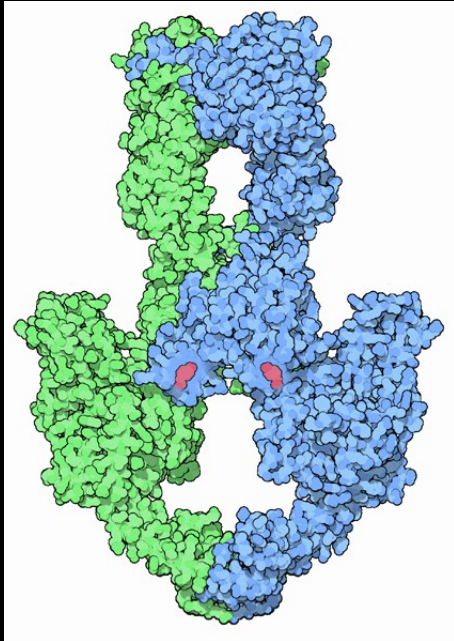
Topoisomerase II

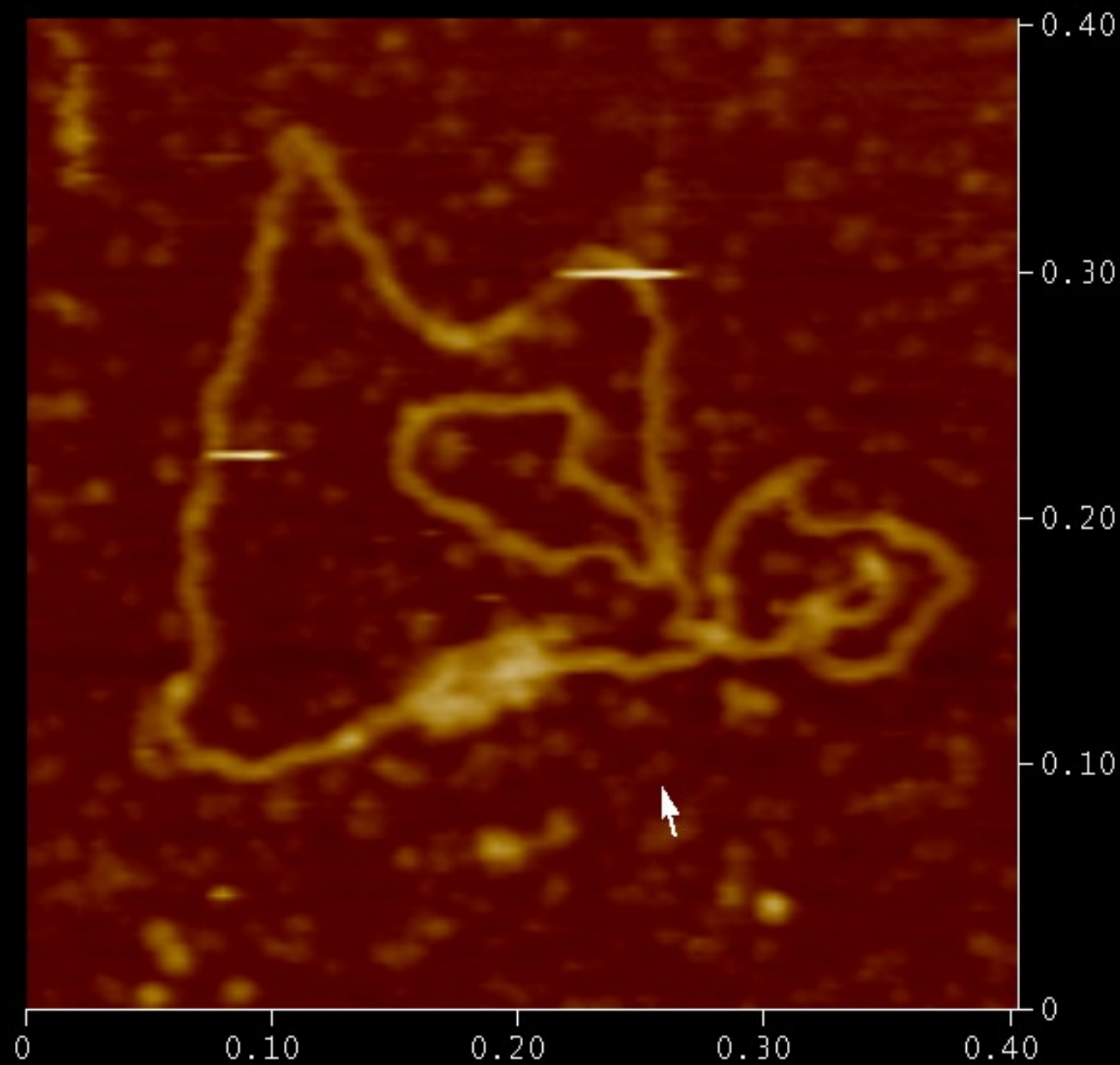


Mecanisme d'action de la Topo II

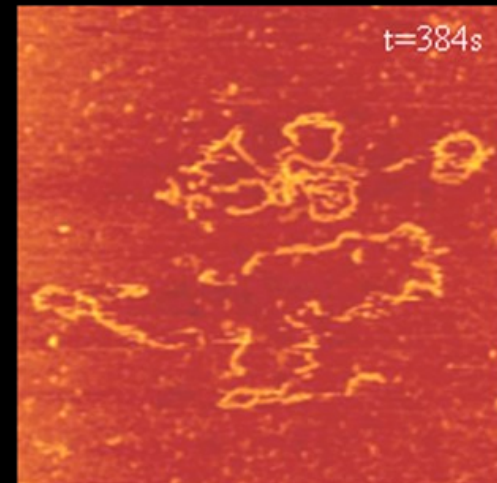
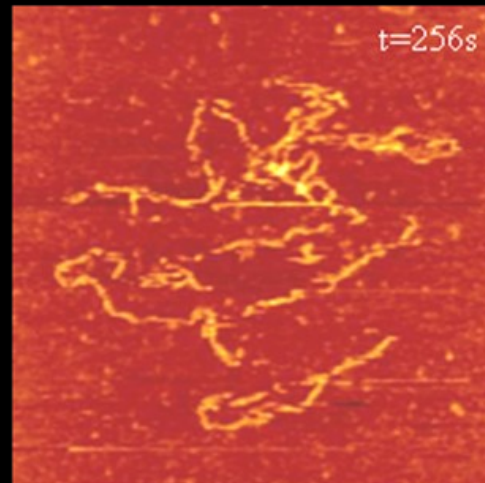
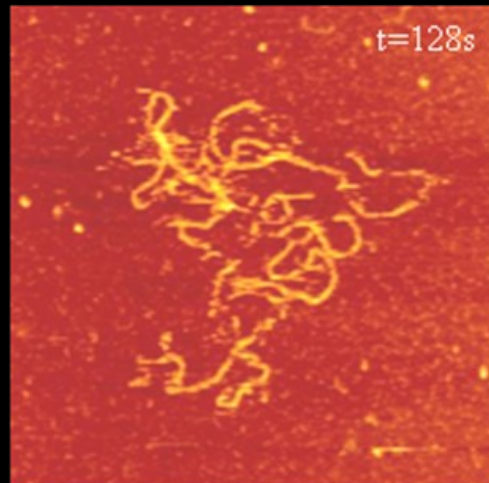
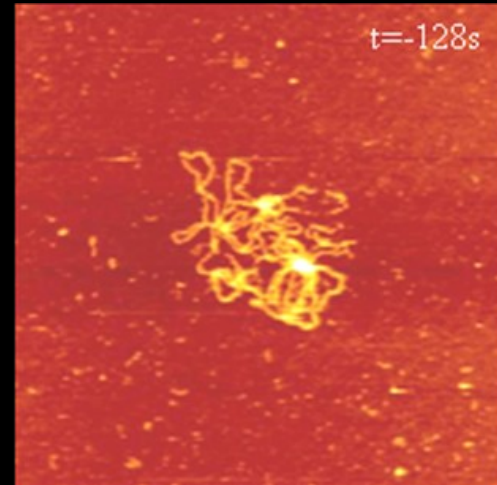
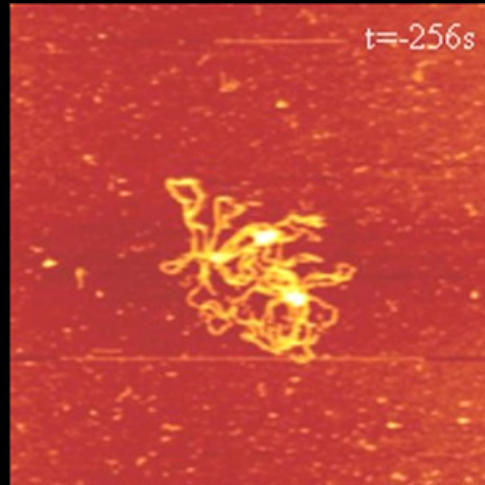
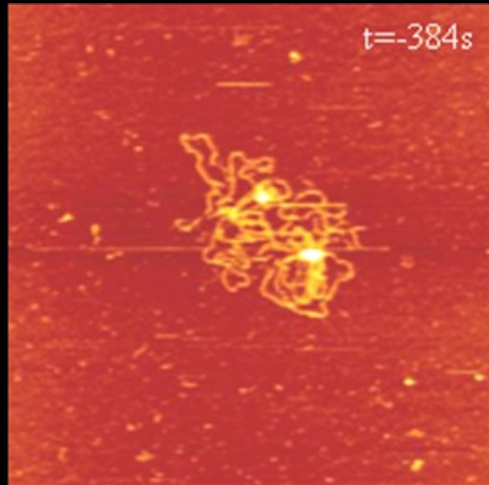


Structure de la Topo II

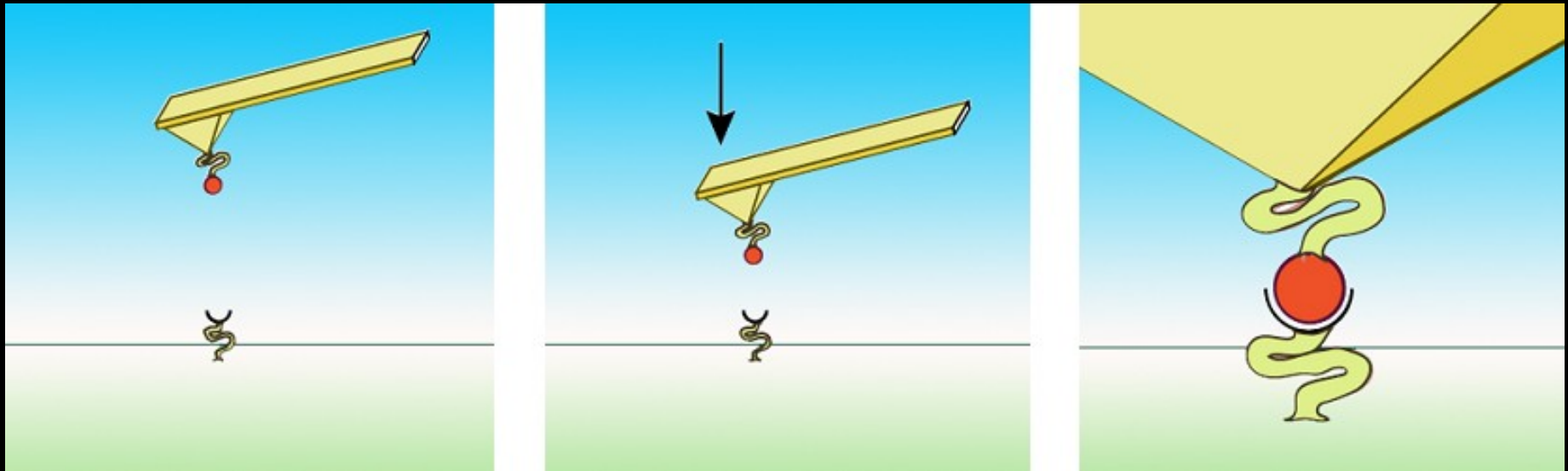




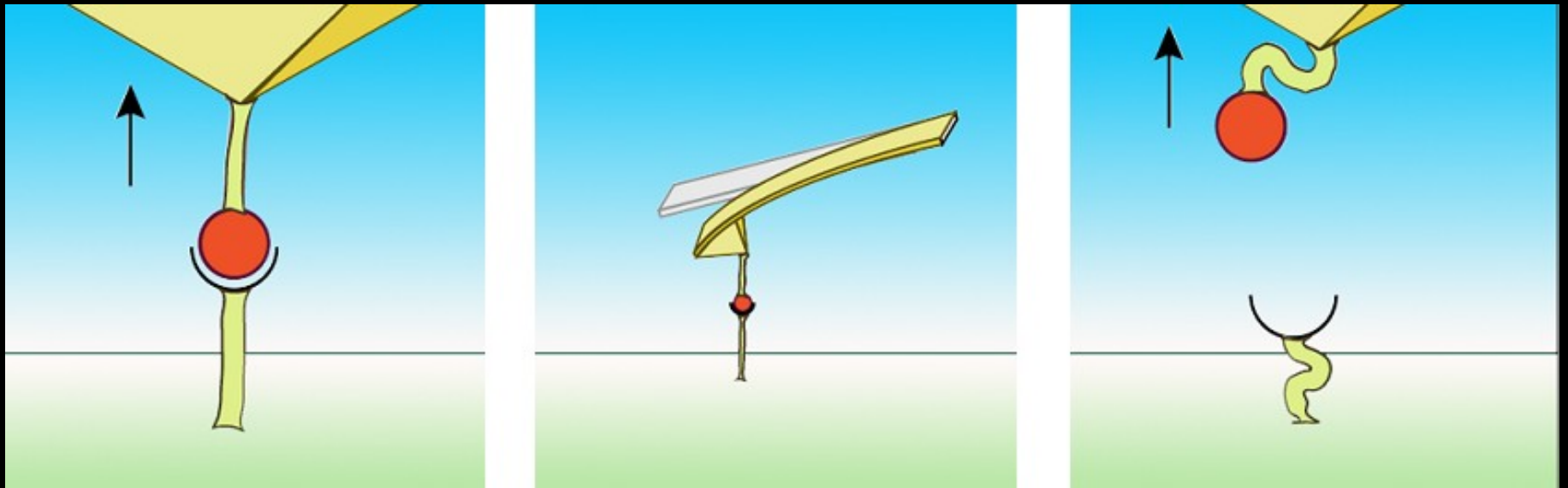
Activité de la Topo II sur l'ADN



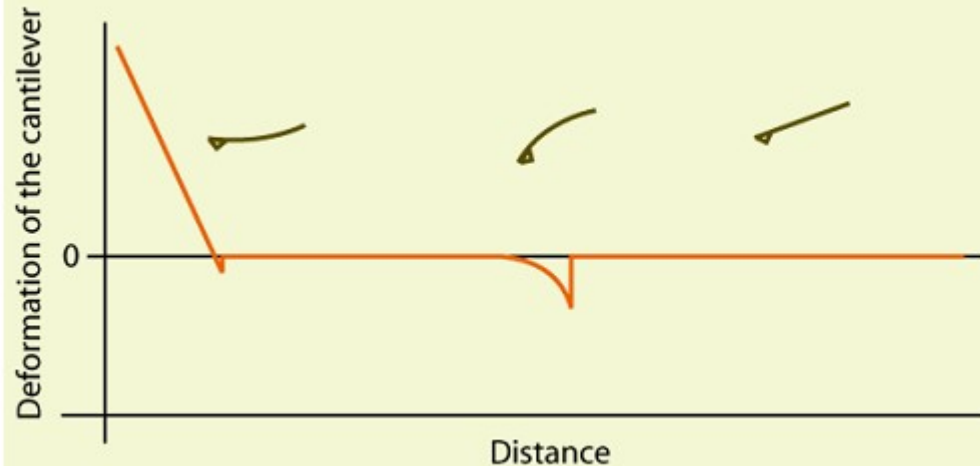
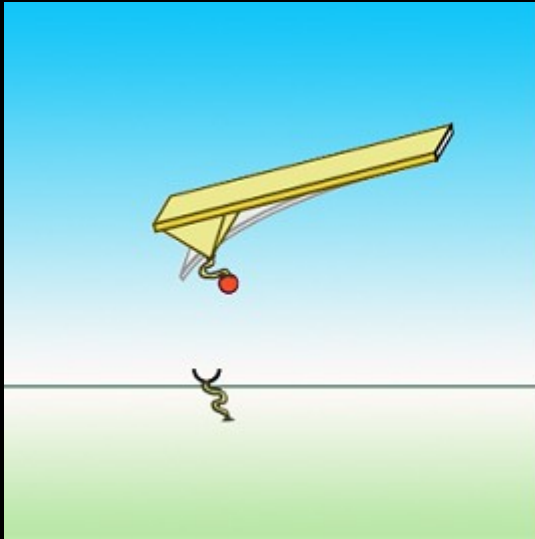
Mesure des forces d'interaction entre protéines



Mesure des forces d'interaction entre protéines



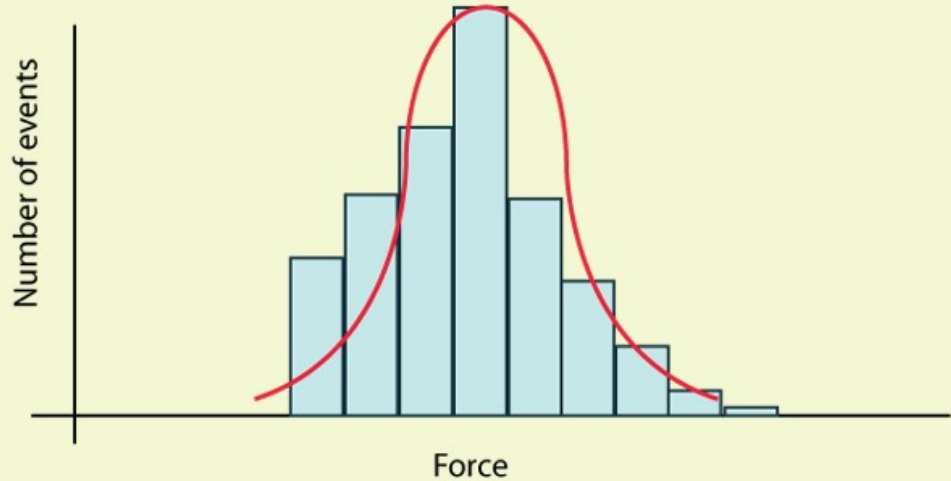
Mesure des forces d'interaction entre protéines



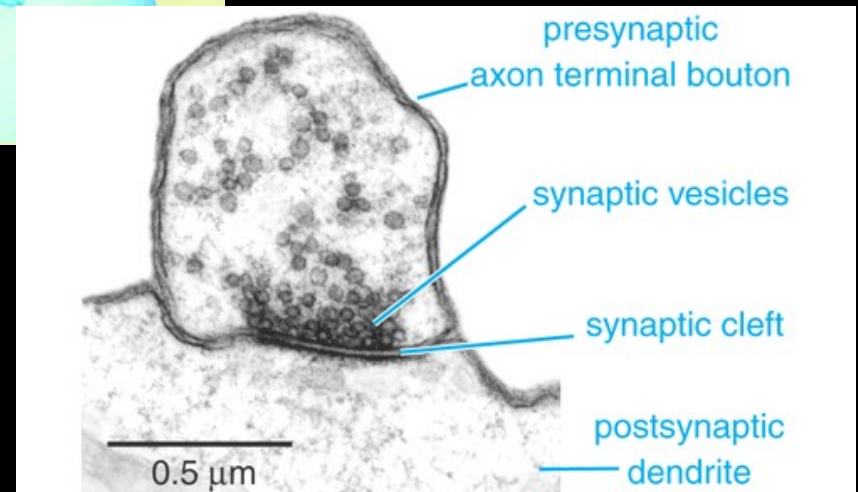
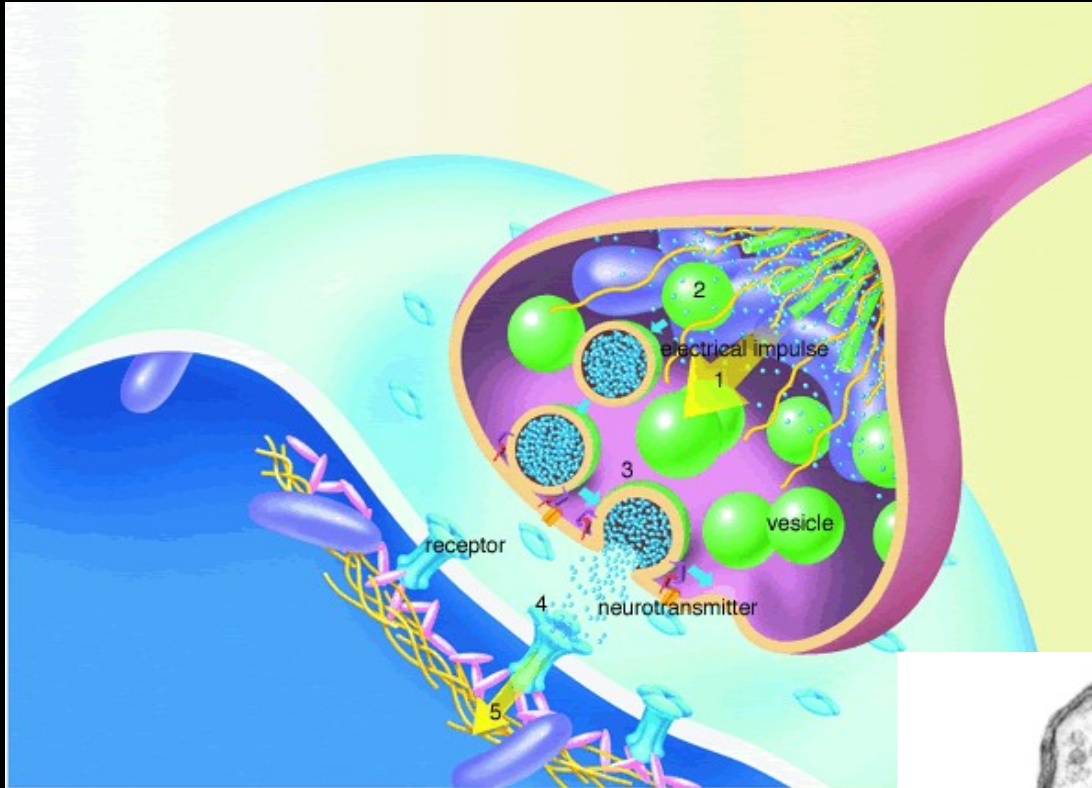
Mesure des forces d'interaction entre protéines



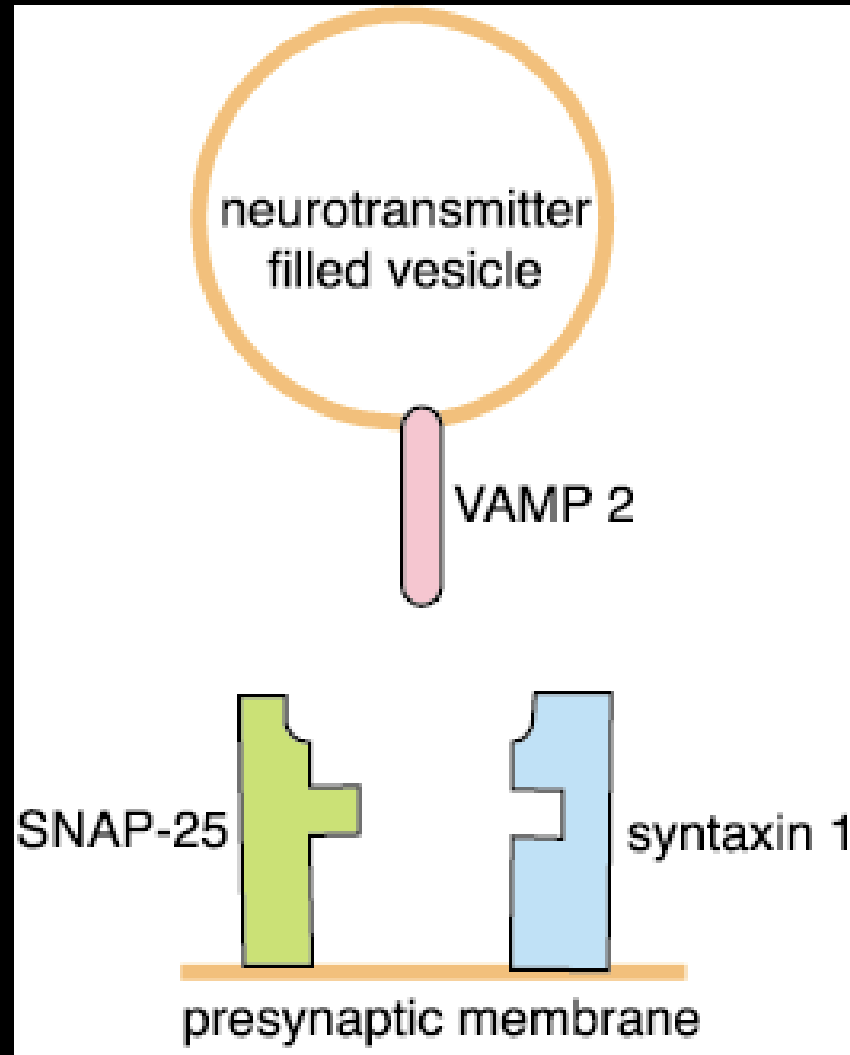
$$F = k * dX$$



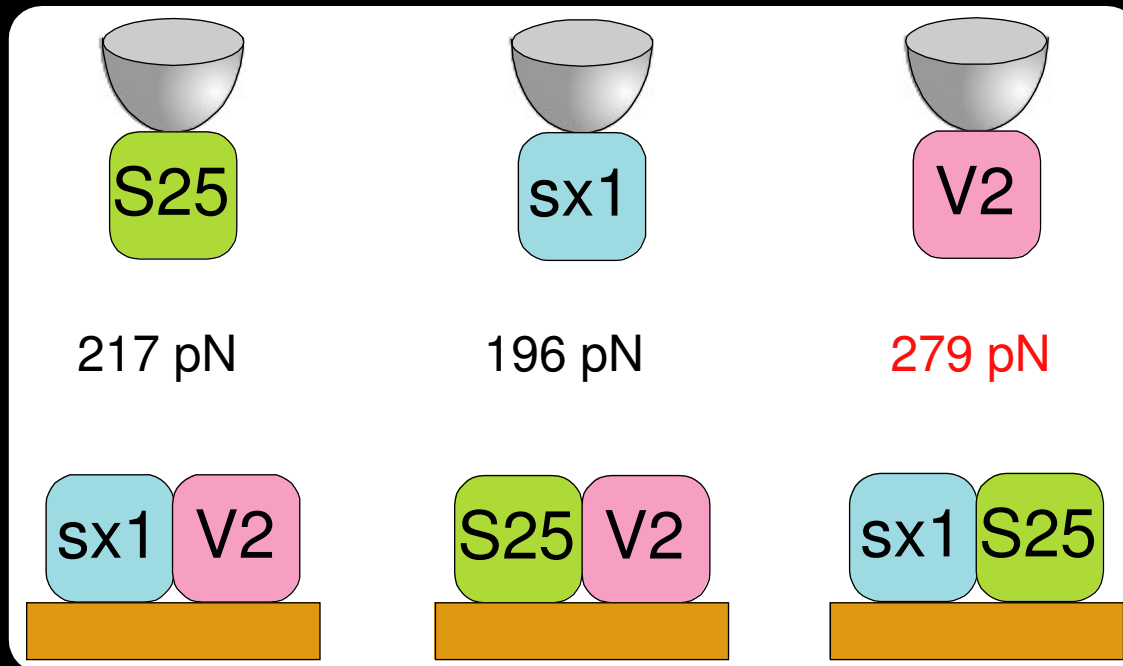
Le complexe SNARE



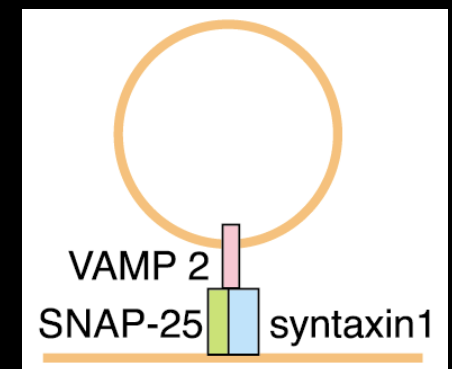
Complex SNARE



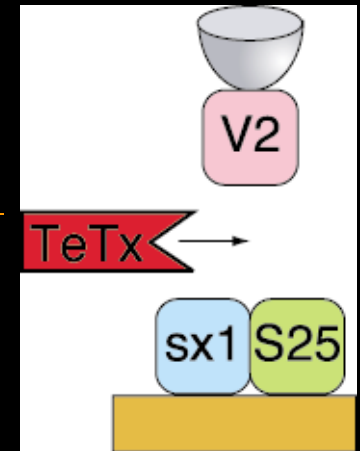
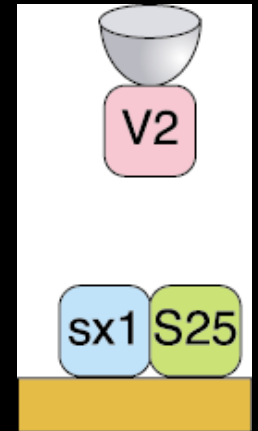
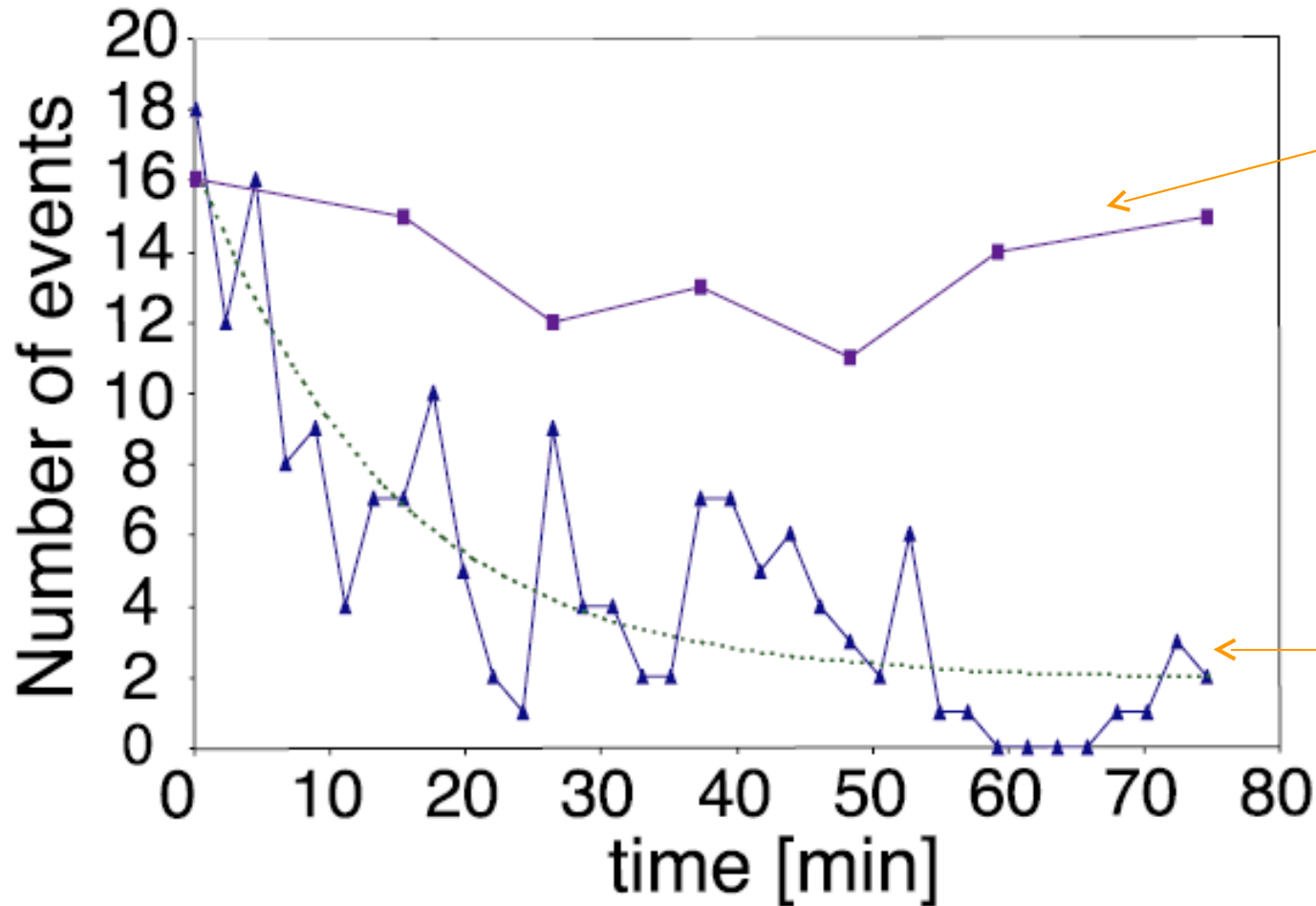
Deux protéines pré-mixées sur le mica

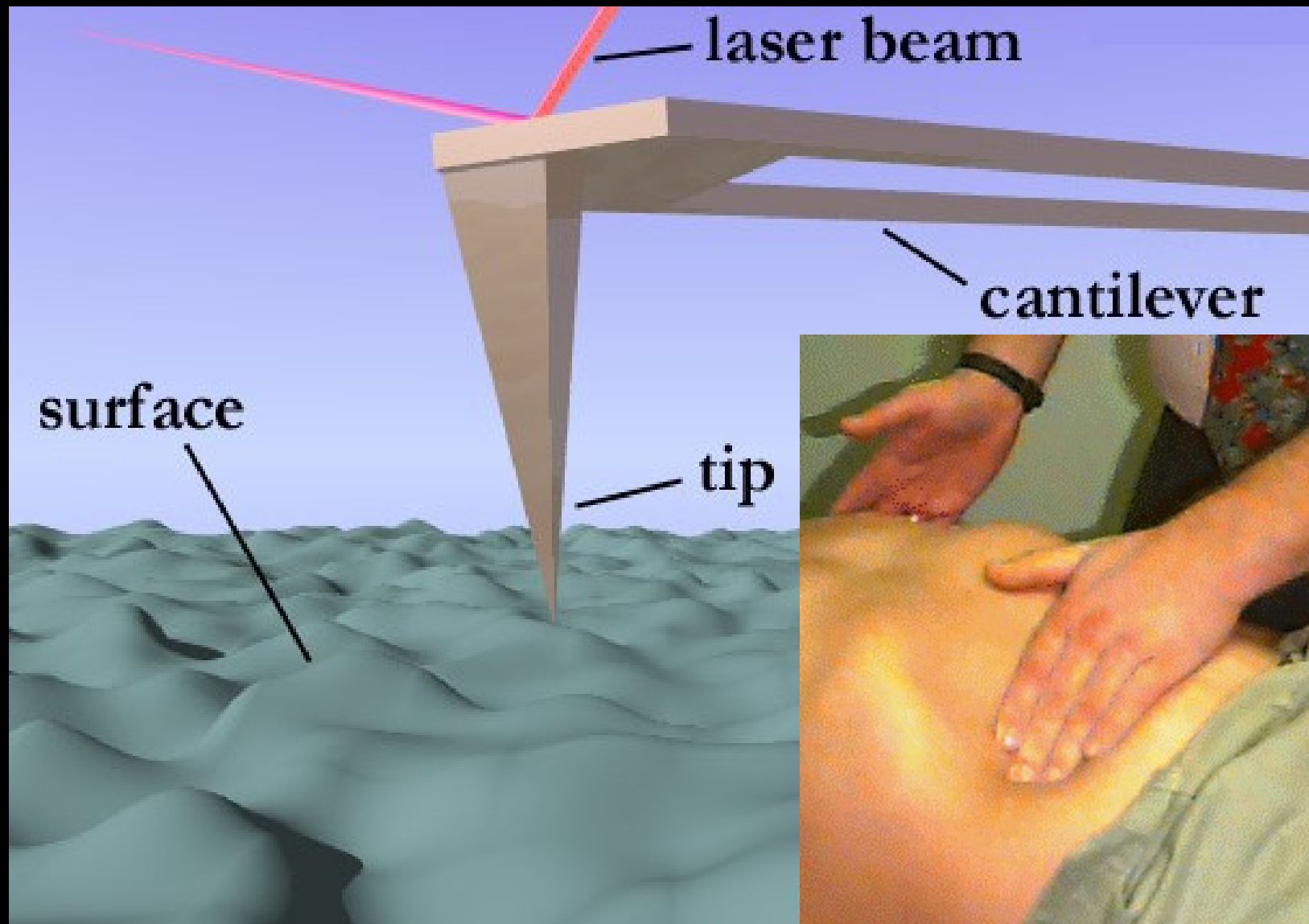


L'interaction la plus forte correspond à la configuration physiologique

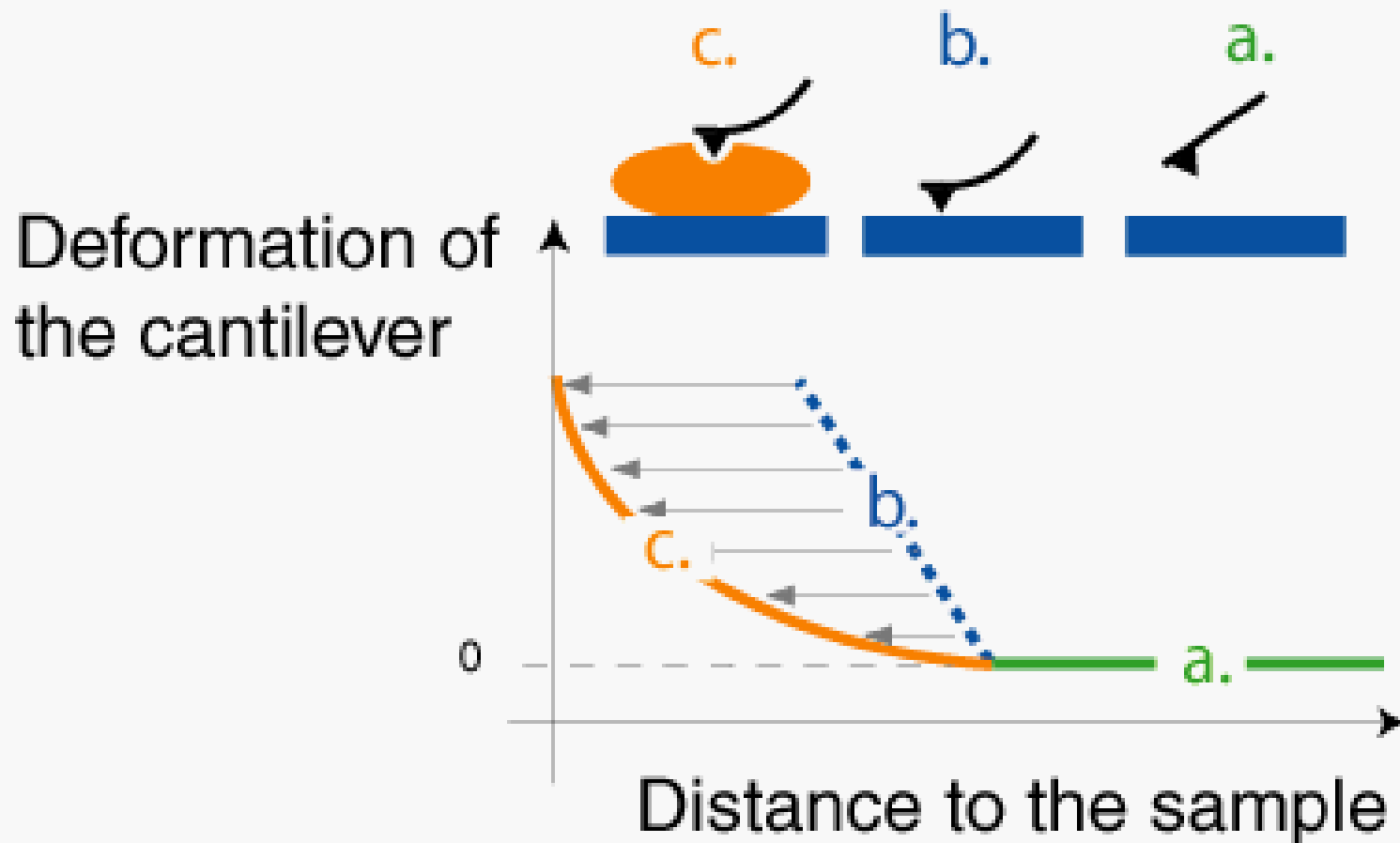


Effet de la toxine tétanique

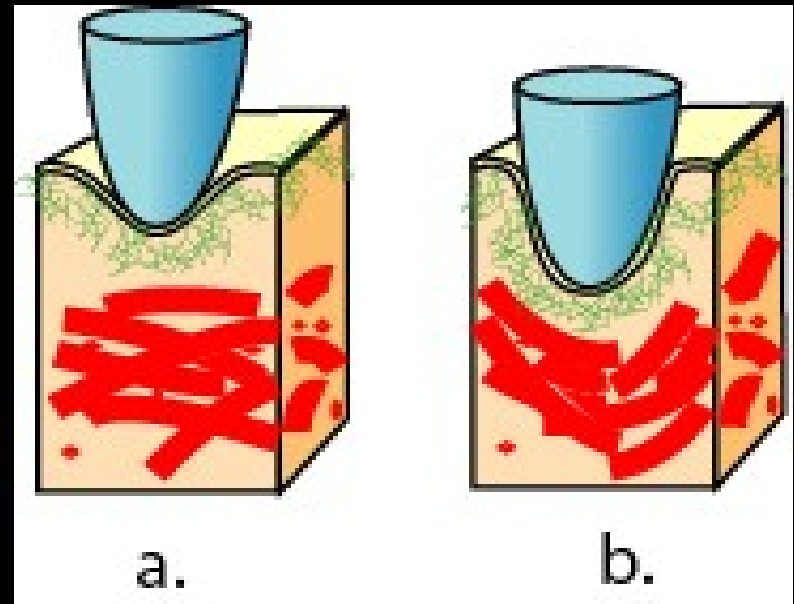
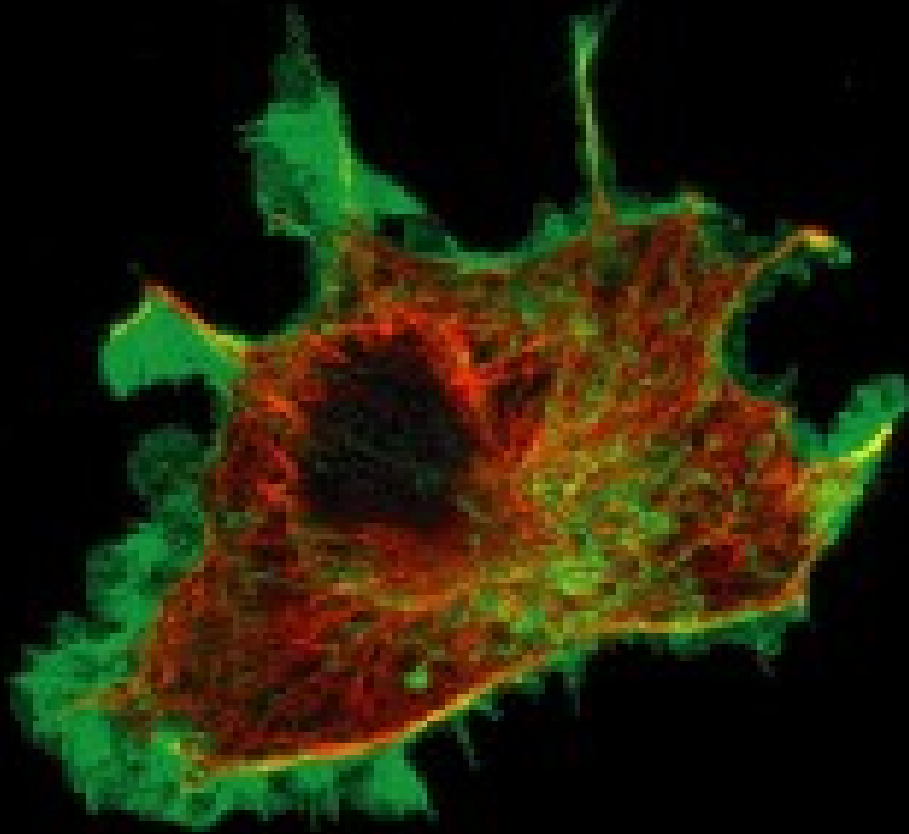




Indentation



Membrane et cytosquelette



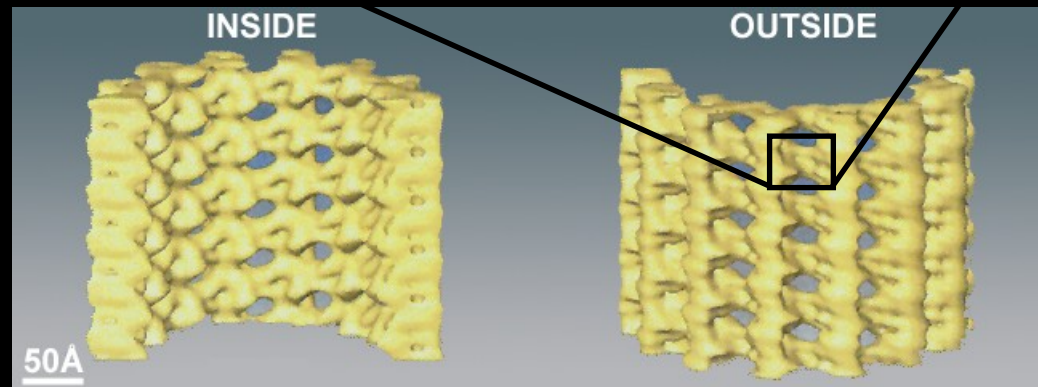
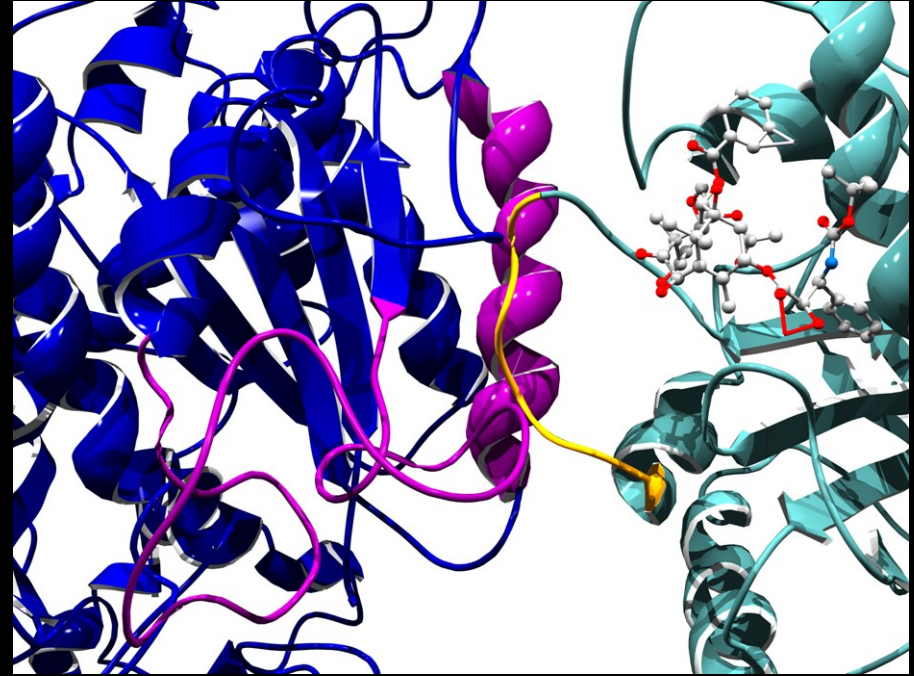
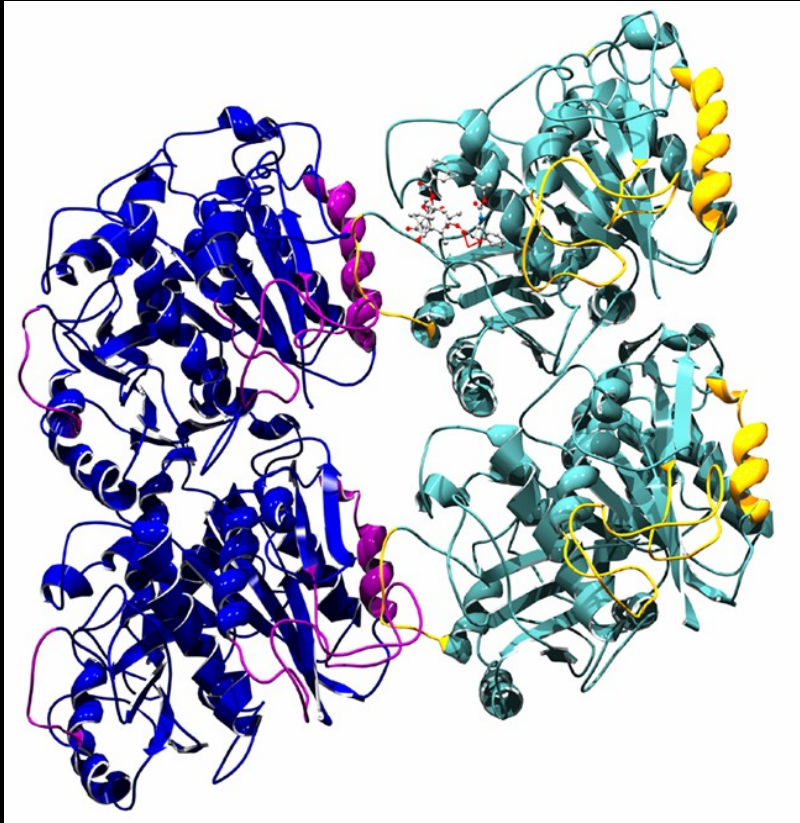
Vert :

actine

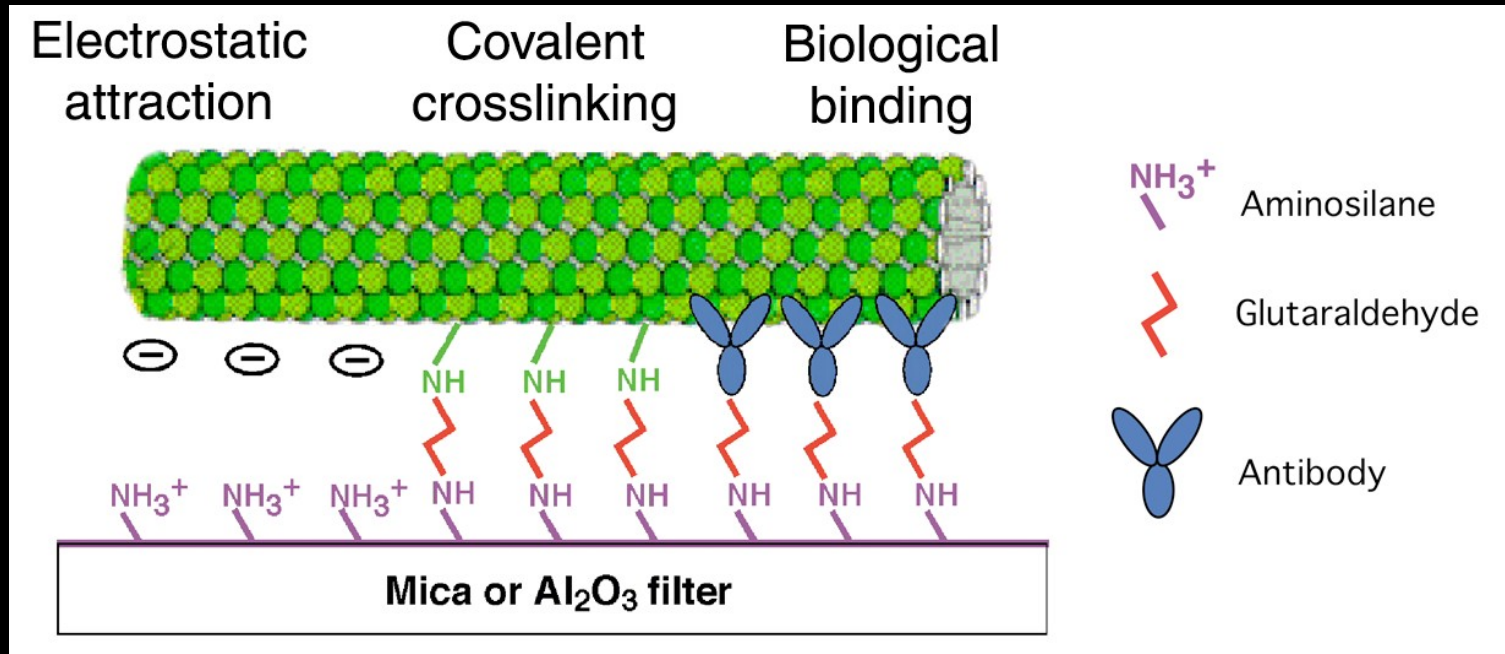
Rouge:

tubuline

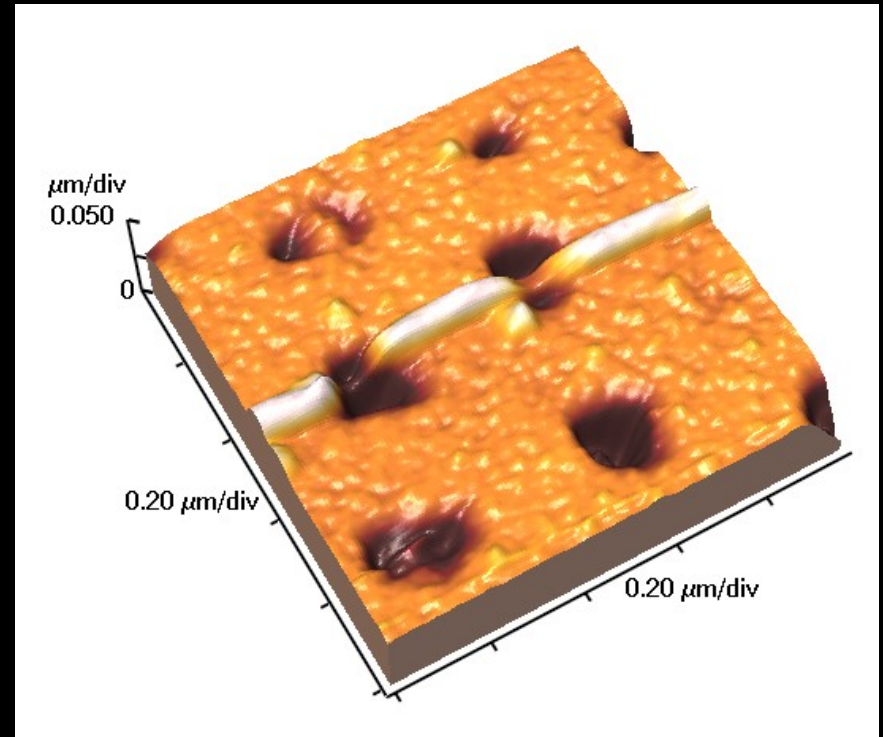
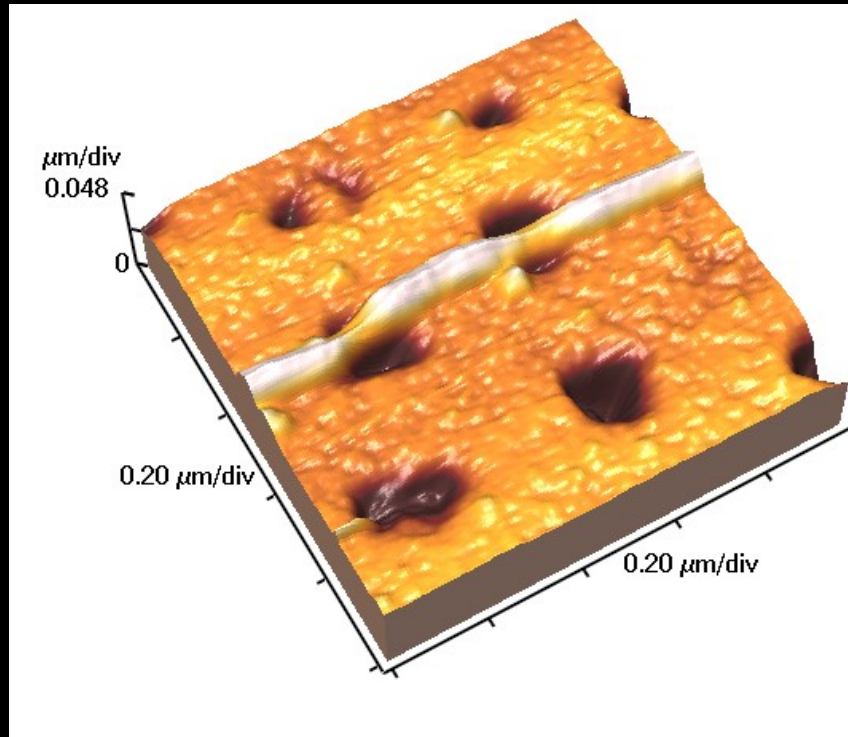
Structure des MT



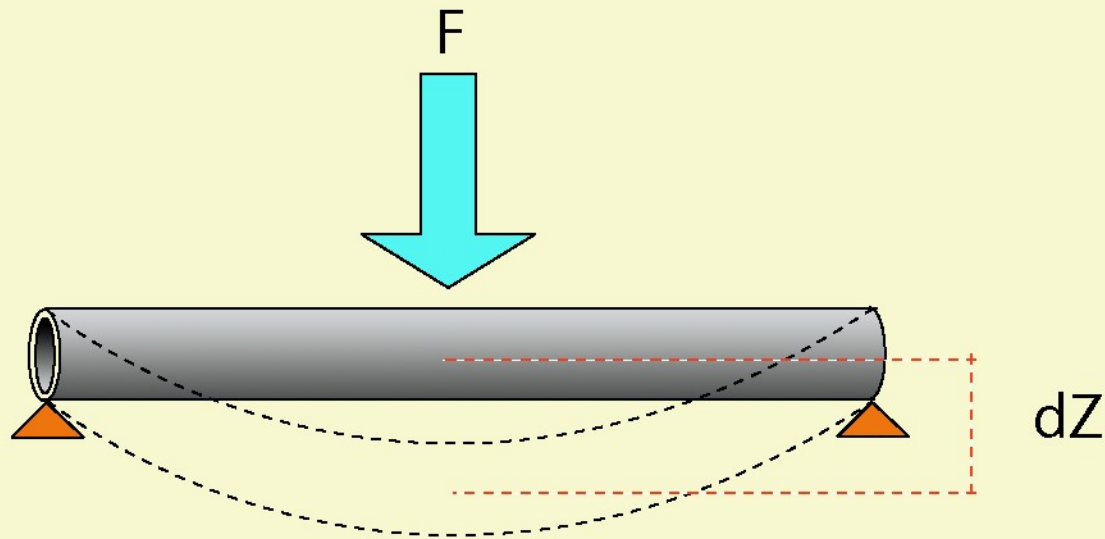
Fixation des MT sur le substrat



Déformation verticale des MT

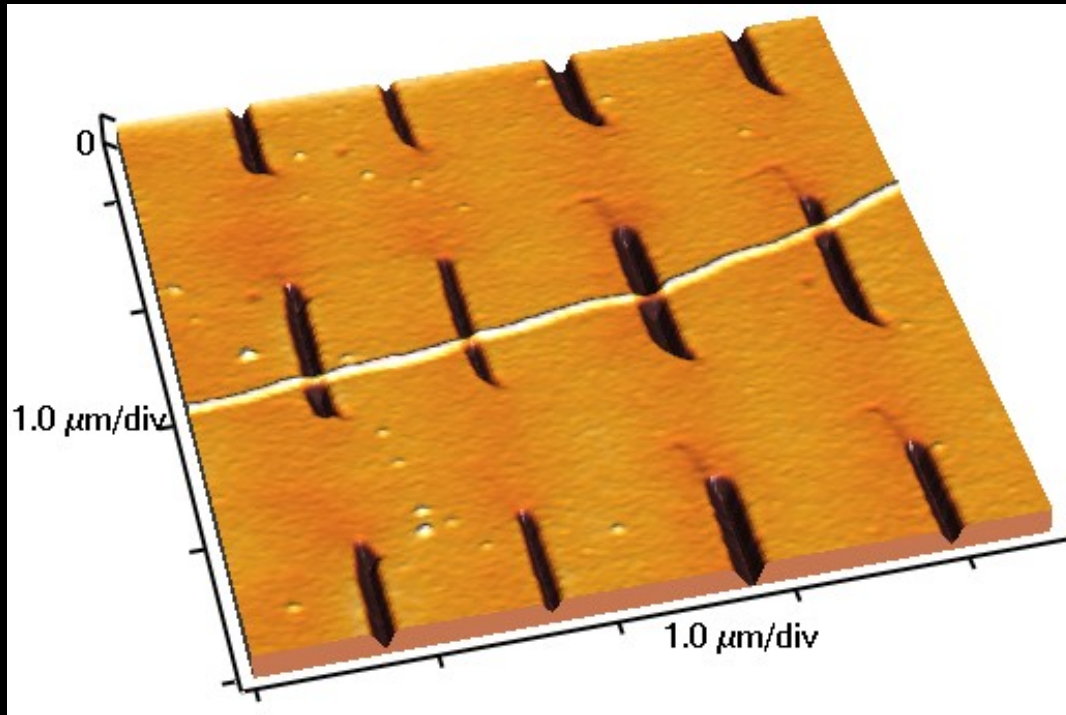


Le model mathématique



$$dZ = \frac{F * L^3}{192 * E * \pi * (D_{\text{ext}}^4 - D_{\text{int}}^4)}$$

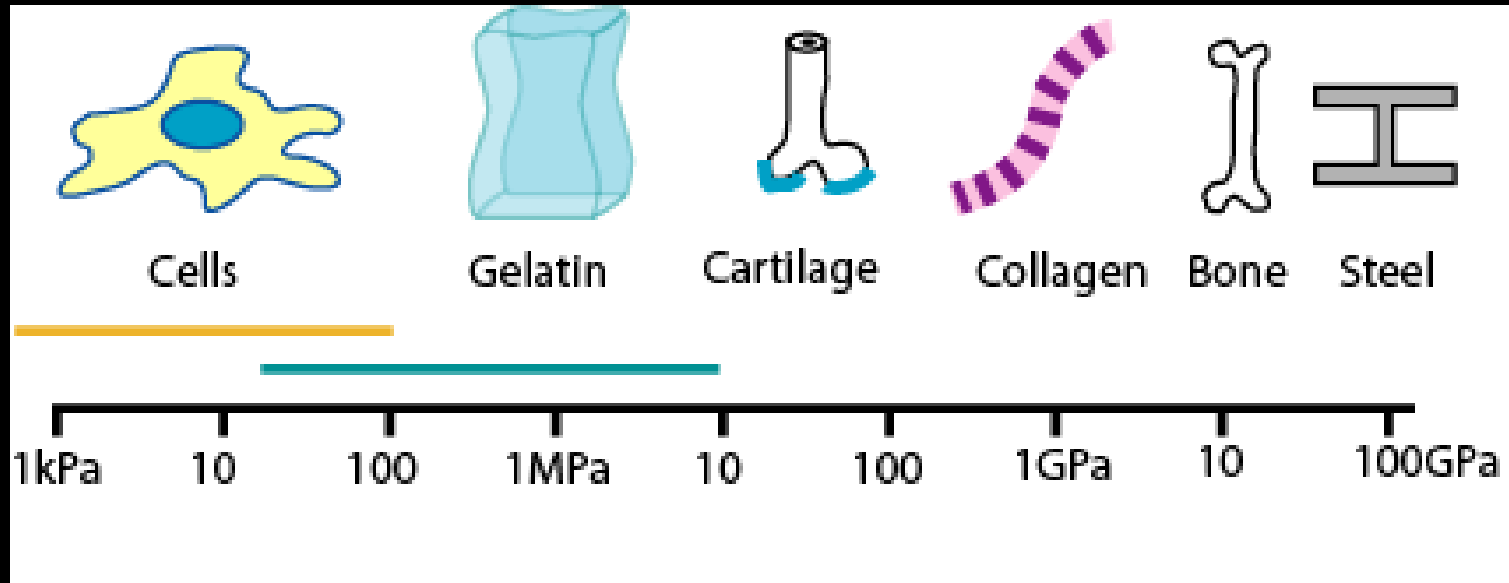
Séparation du module de Young de celui du cisaillement



$$G = (1.5 \pm 0.3) \text{ MPa}$$

$$E_{\text{Young}} > 130 \text{ MPa}$$

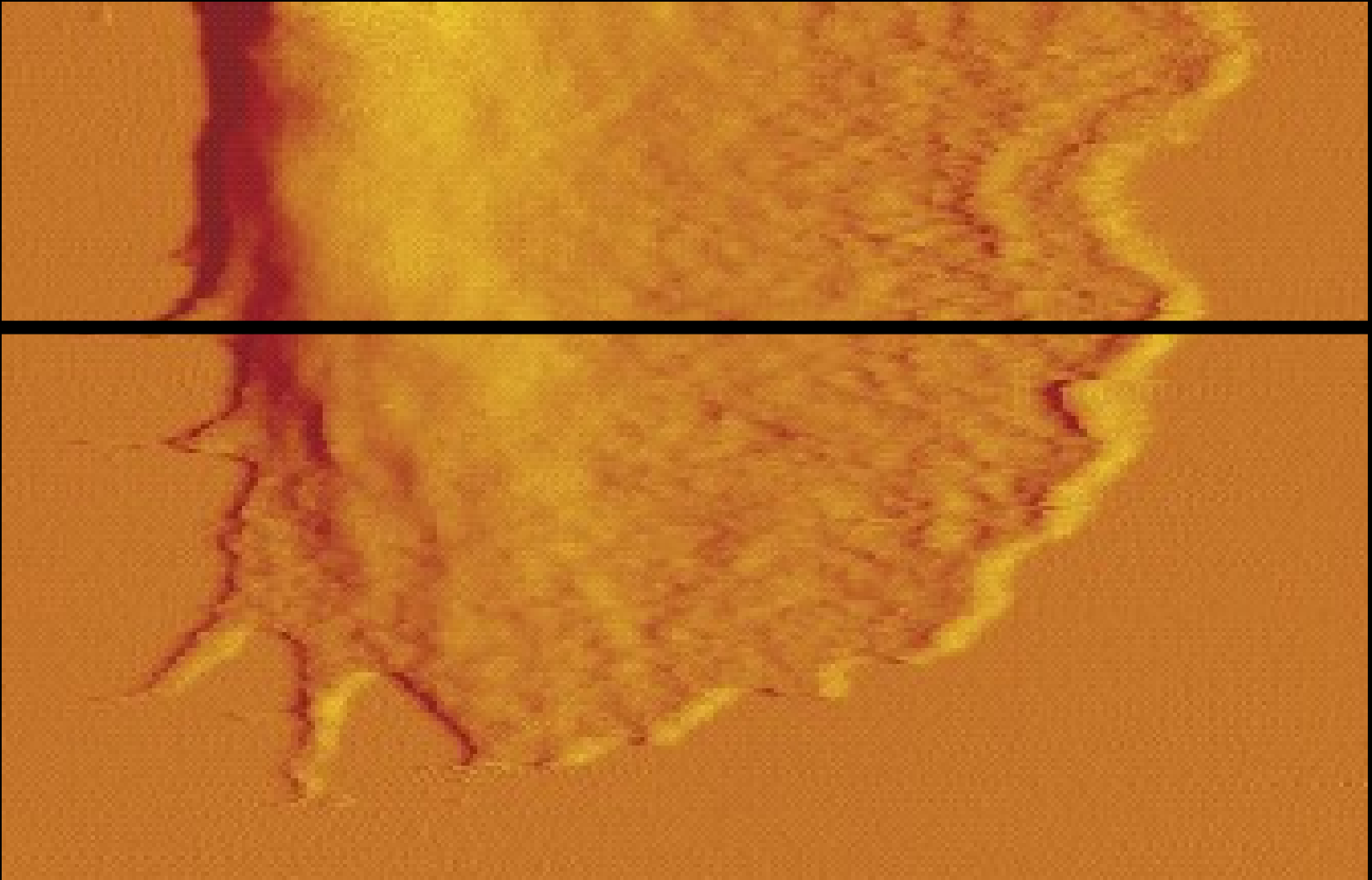
Young's modulus



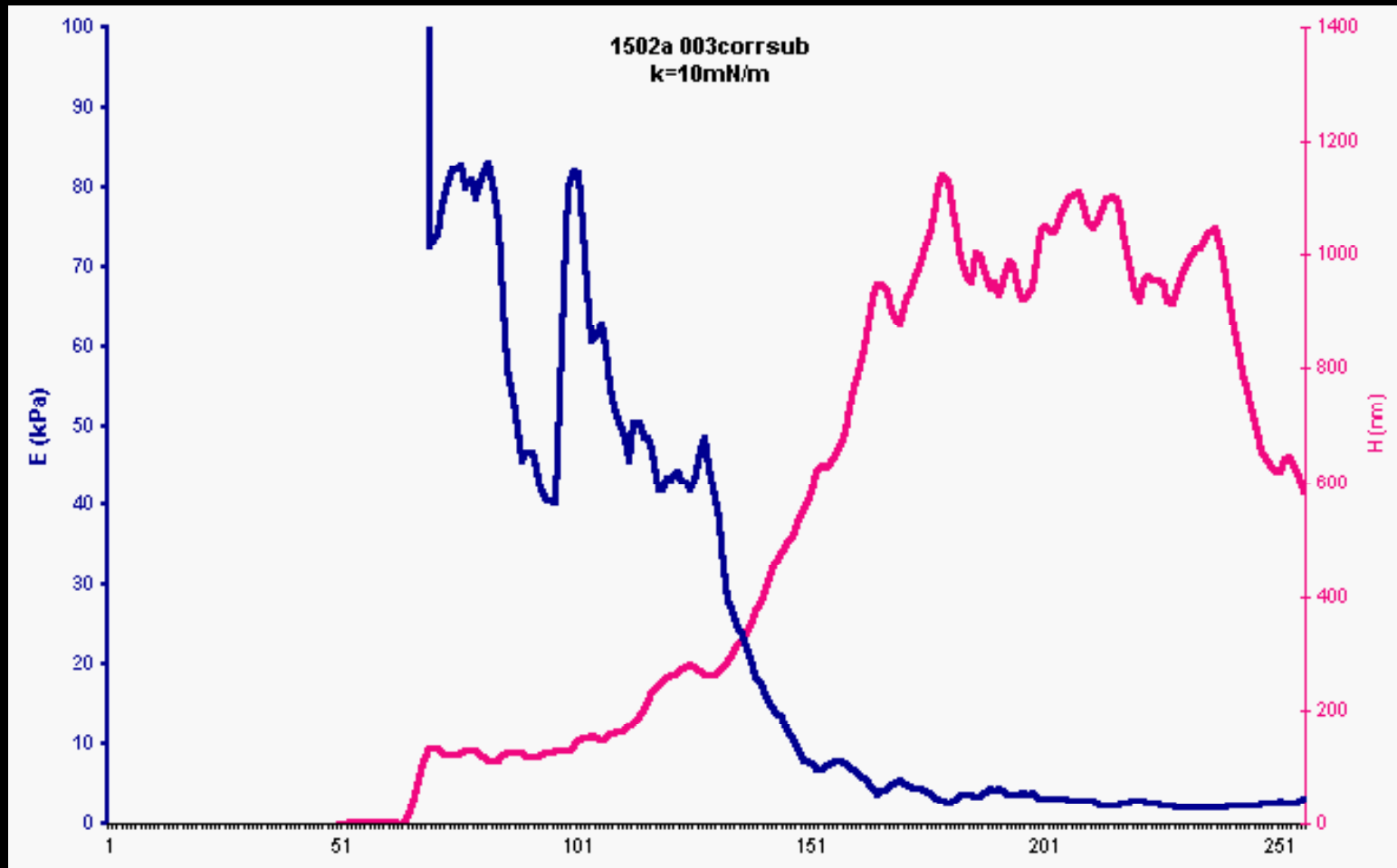
Keratocytes

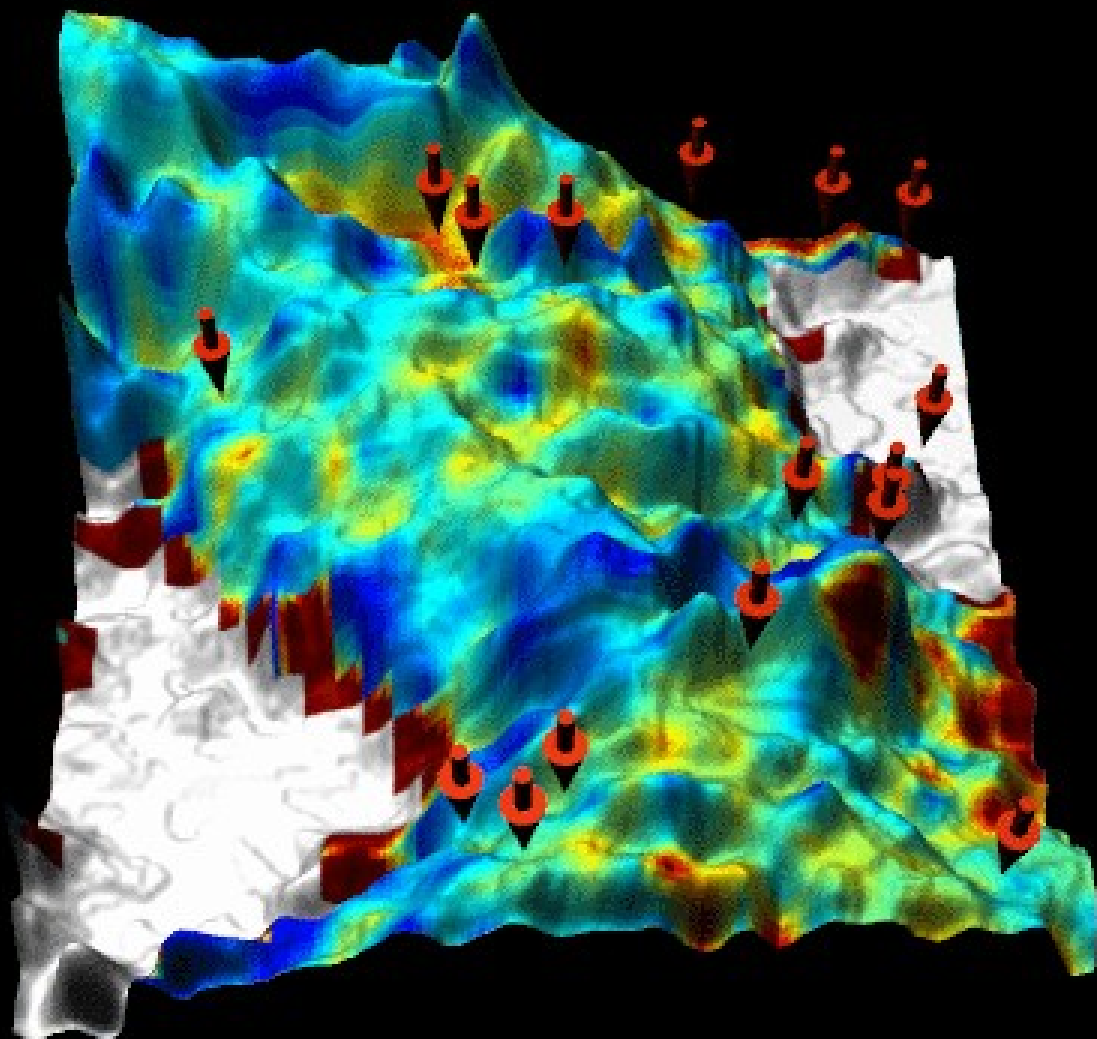


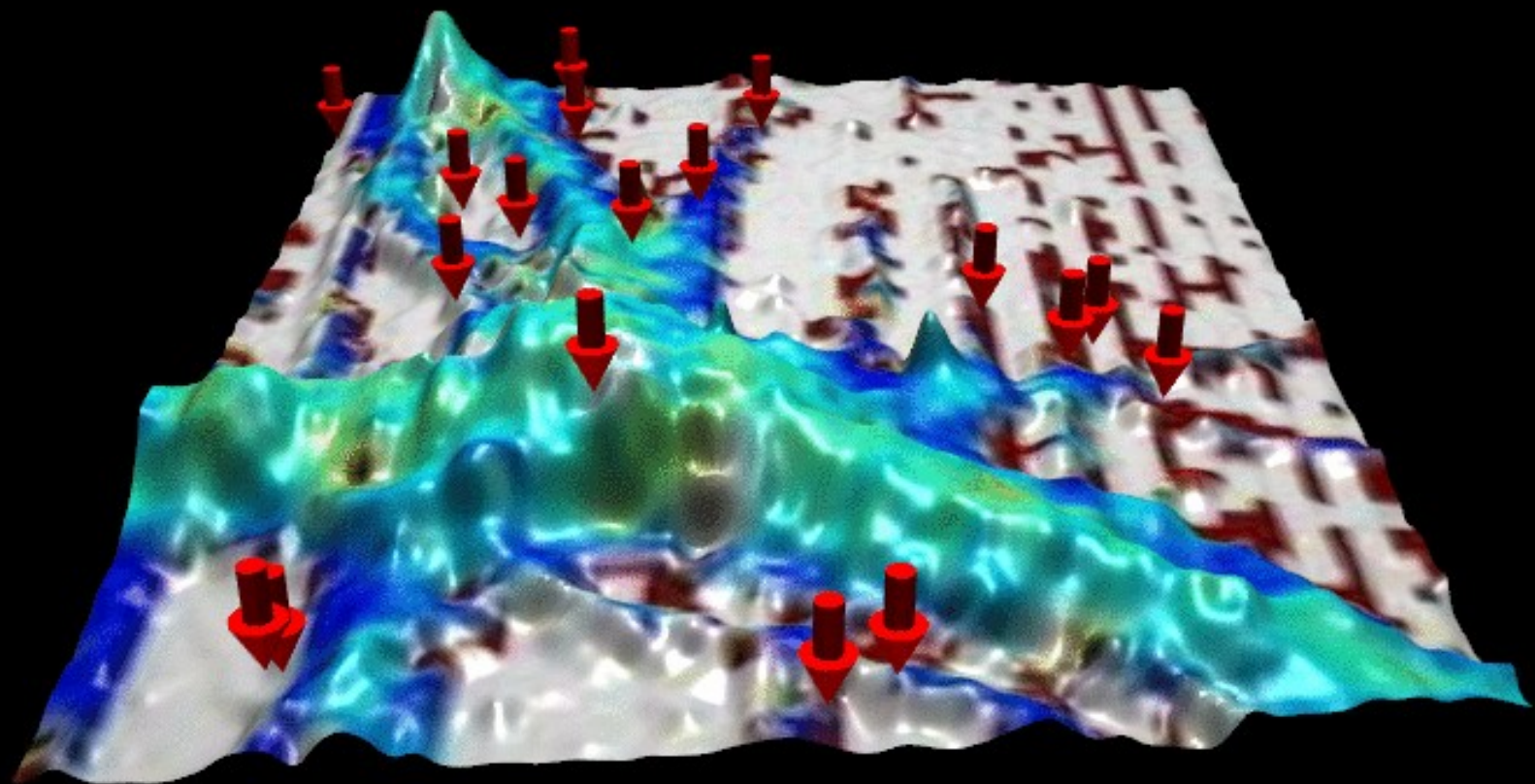
Lamelipode



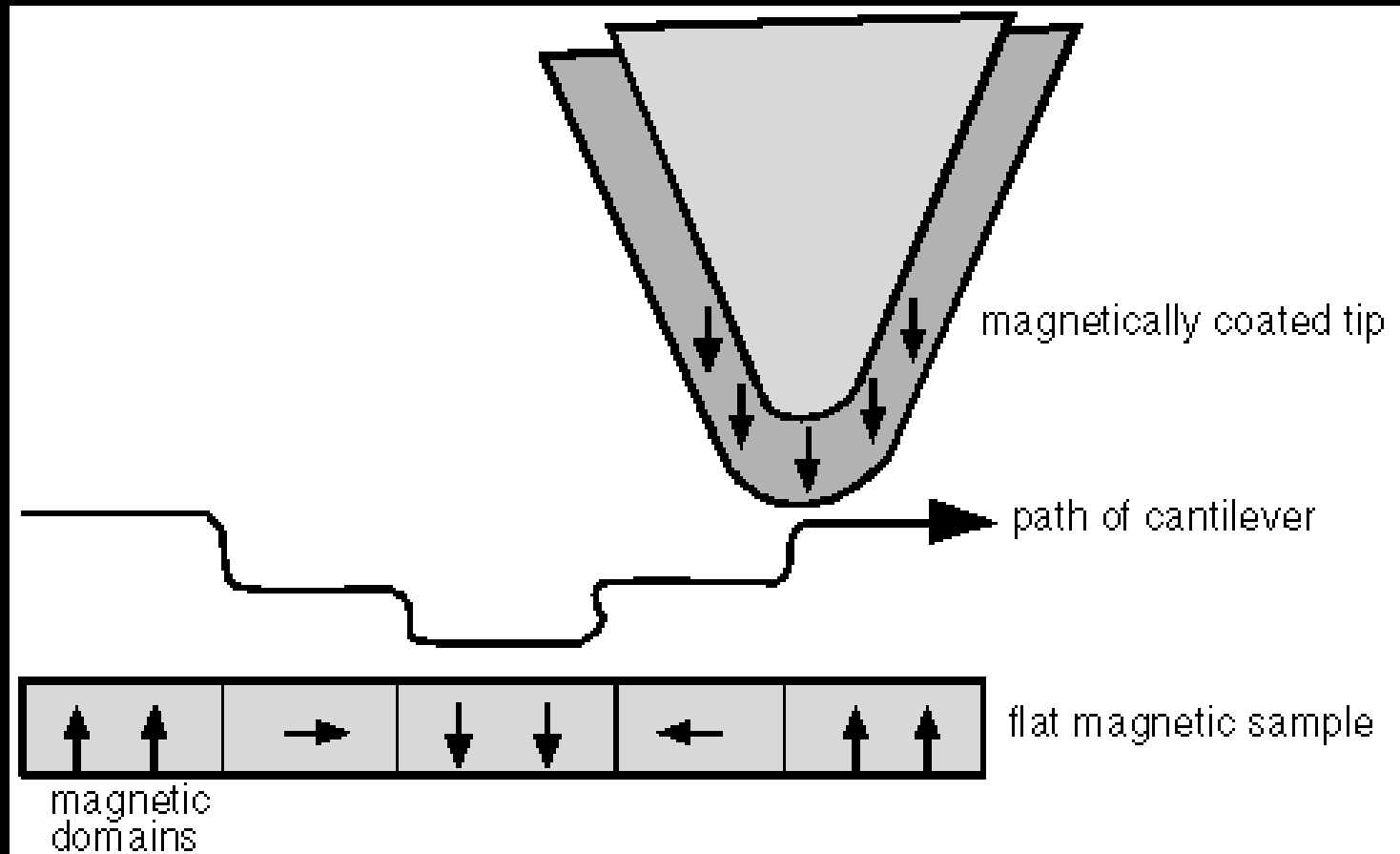
Profil d'élasticité des kératocytes



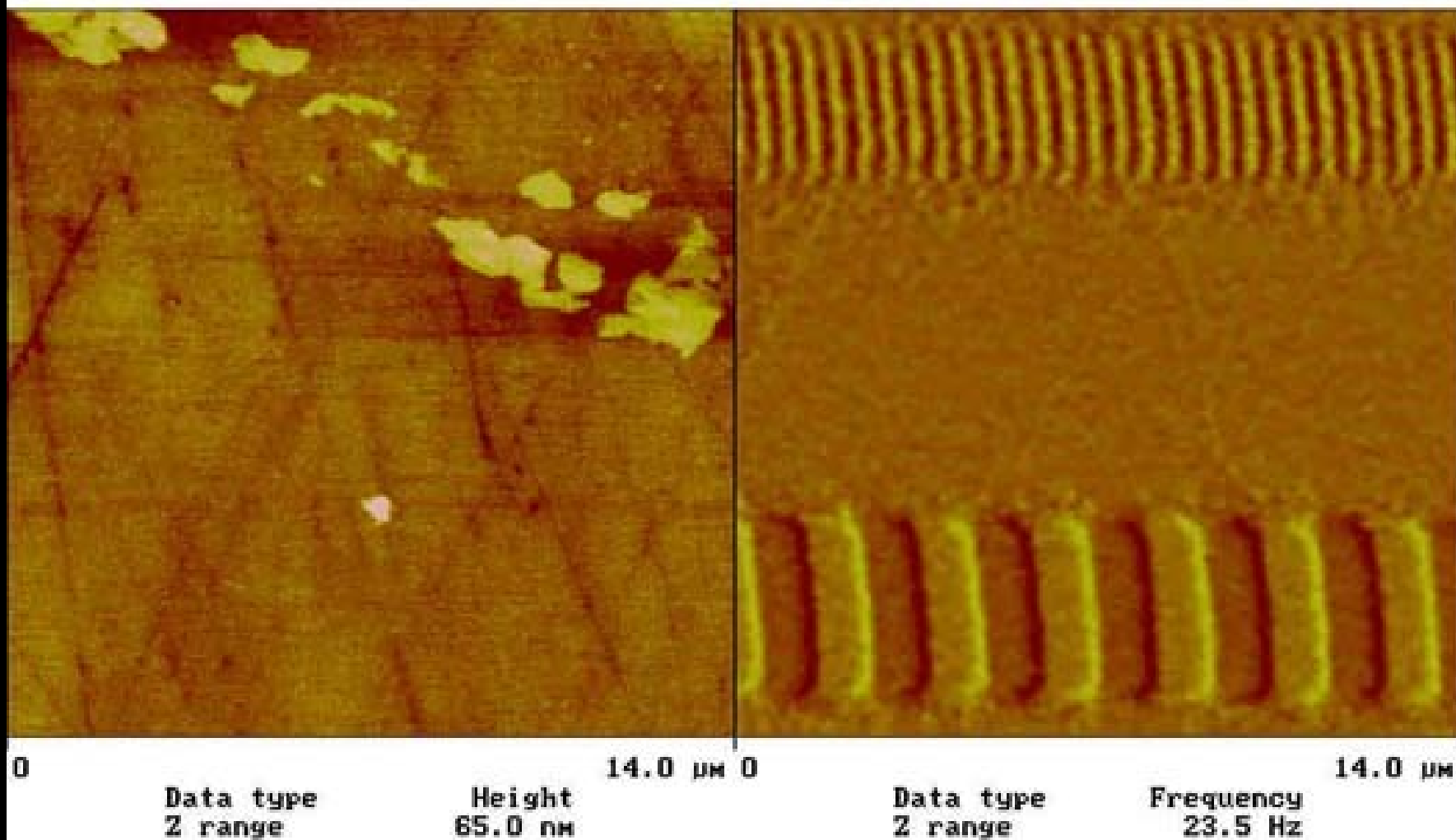




Magnetic Force Microscopy



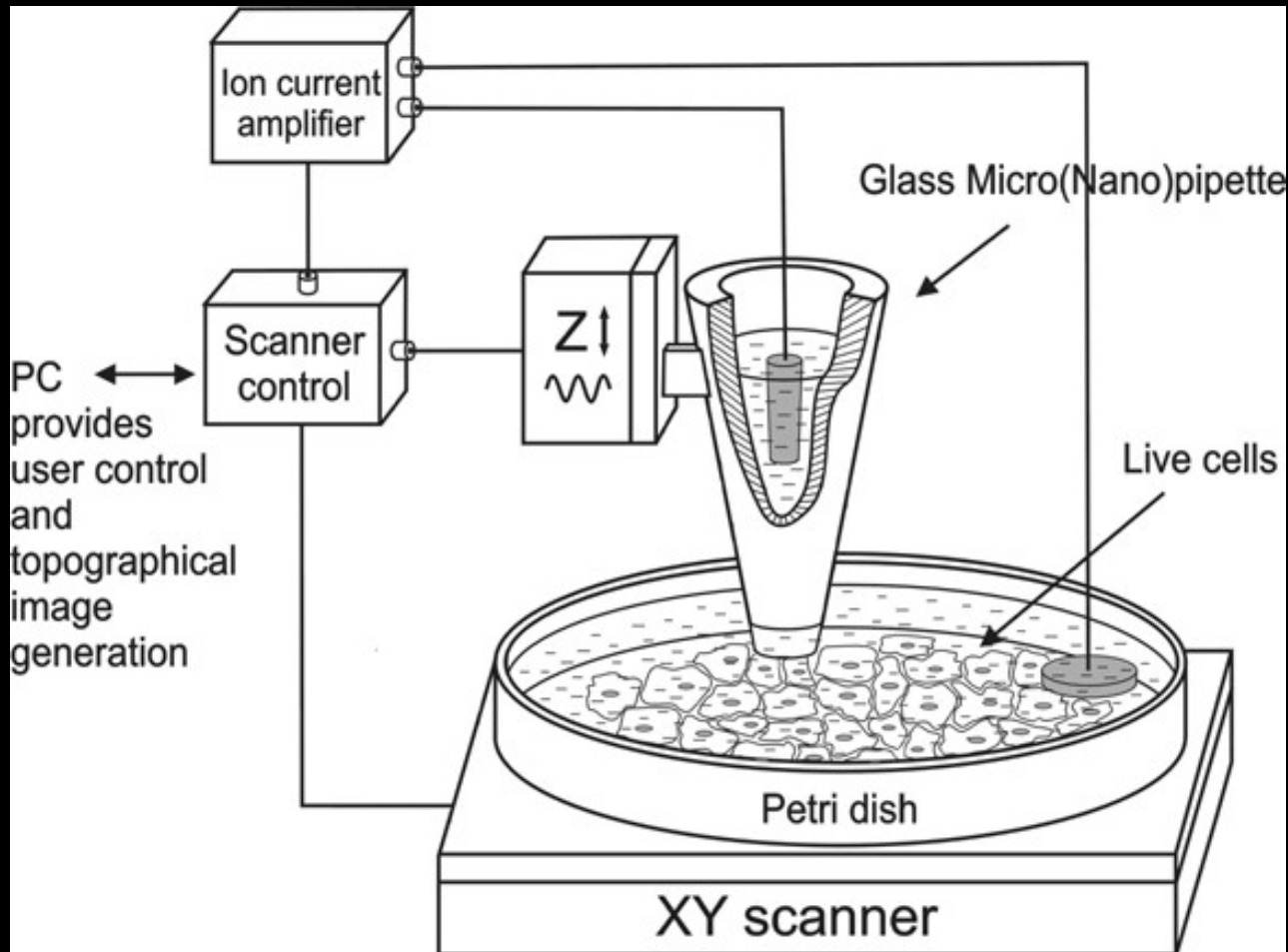
XYDMAFM Images



mfmf32_2.004

Sample: Exp.Disc, IBM; Probe: Nanosensors, MFM

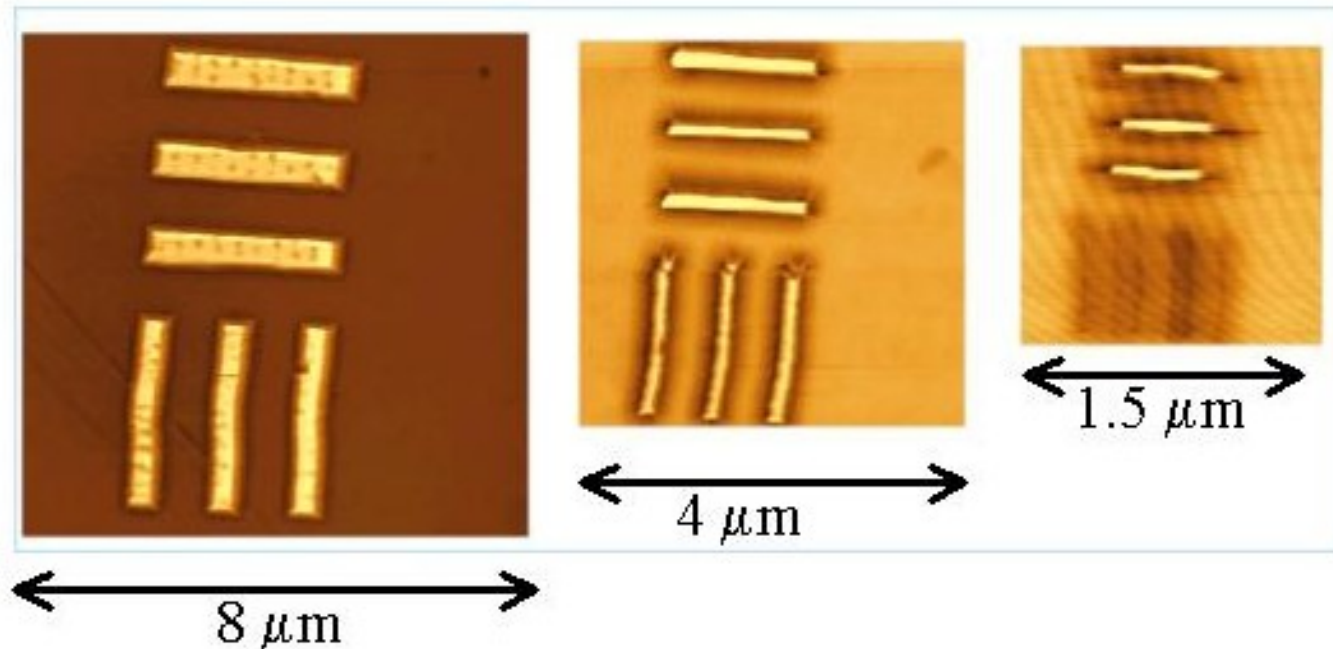
Scanning Ion Conductance Microscope



Scanning Thermal Microscopy



Scanning thermal microscopy



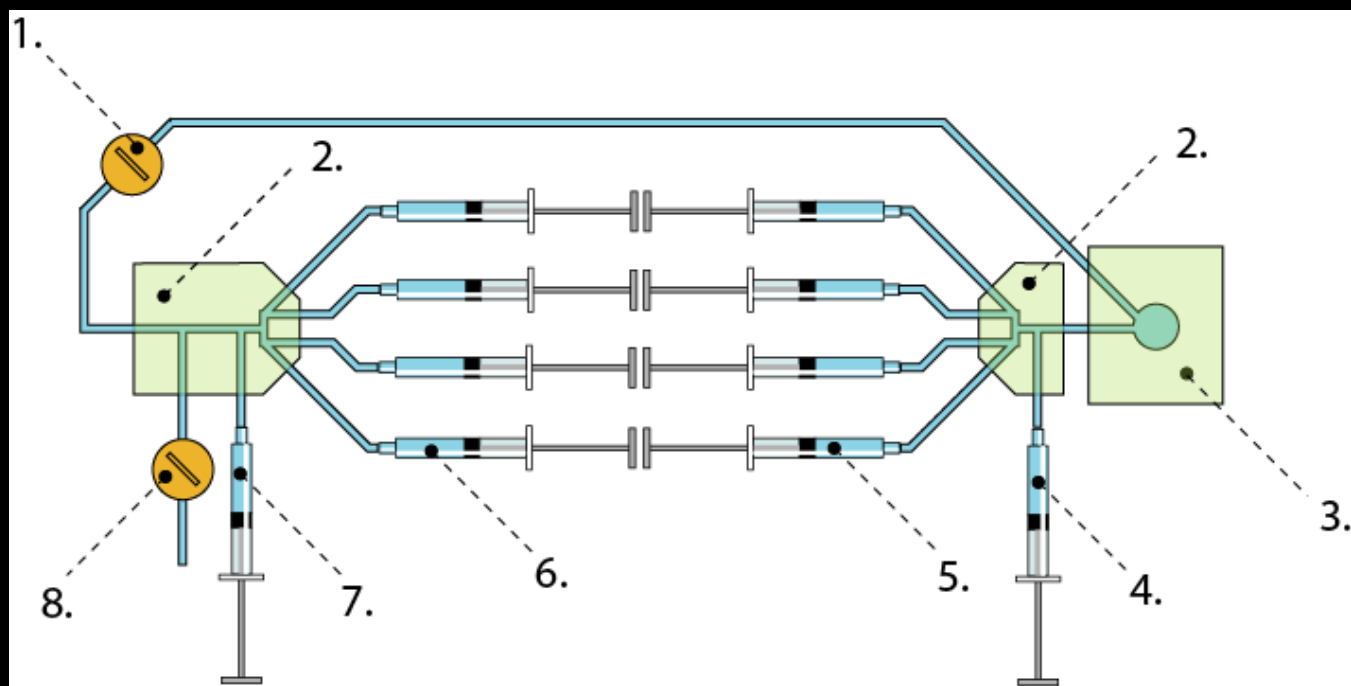
Bowtie probe DC thermal images of a model sample consisting of silica coated with a discontinuous layer of PMMA of thickness 500 nm (sample prepared using photolithography, by Dr G Mills, Glasgow University; imaging by Dr A Hammiche, Lancaster University). Gaps in the PMMA appear as lines, exposing the silica which has higher thermal conductivity.

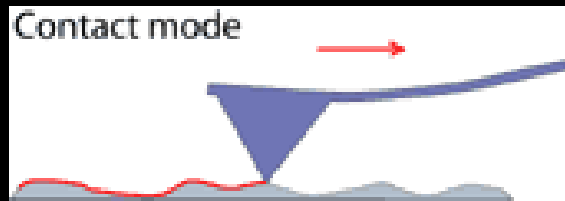
Acknowledgments



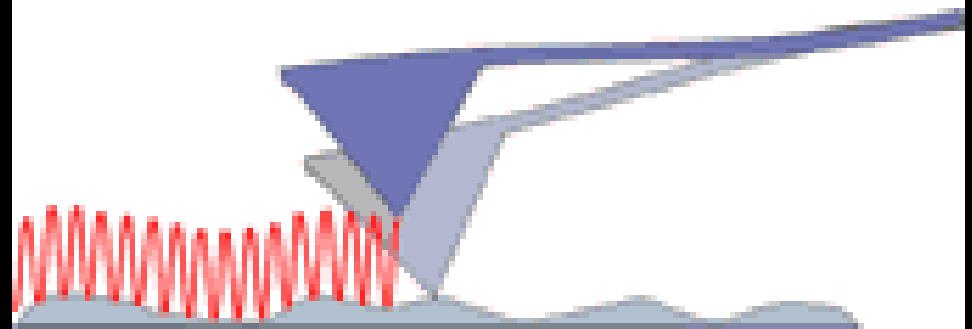
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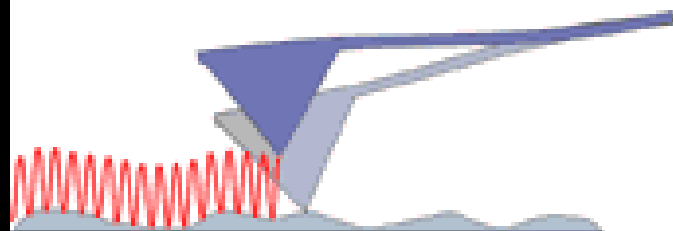




Intermittent contact



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Non-contact mode

